

Generating biophysical-simulator parameters from observed
behaviour alone:
a general route past the invasive-measurement bottleneck,
demonstrated across worm and fly

Chenxi He
Cavendish Laboratory
University of Cambridge
Cambridge CB3 0HE, United Kingdom
ch2067@cam.ac.uk

June 21, 2026

Plain-language summary. Whole-organism computer simulators of the worm *C. elegans* and the fly *Drosophila* let researchers test how brains and bodies produce behaviour. But each simulator carries thousands of internal numbers (parameters) that normally have to be tuned with invasive electrical recordings that most behaviour laboratories cannot do. We show that a laboratory only needs ordinary tracking video of an animal: a procedure called **PGOB** takes that video and returns a calibrated simulator, plus a map of which parameters explain any remaining gap between the simulator and the real animal. We apply the same procedure – one loss, one behavioural evaluator and the same inference engines, with a dimension-adaptive initialisation – to four published simulators across two species and check the generated parameters on biology the procedure never saw, including 43 held-out worm mutant strains and a second worm laboratory. The output is a working simulator and a short list of the specific simulator mechanisms that are still missing.

Abstract

Mechanistic whole-organism simulators turn connectomes and biophysics into testable models of behaviour, but every such simulator is governed by thousands of internal parameters whose values are normally pinned by invasive physiology – intracellular electrophysiology, two-photon imaging, EMG – that most laboratories, and most species, are out of reach of. This calibration bottleneck is general to the field, and it locks out exactly the settings where invasive recording is impossible. We show that the missing ingredient can instead be ordinary tracking video. From the externally observed kinematics of a behaving animal alone, our procedure *generates* a working set of simulator parameters from a start drawn independently of the unknown optimum, and returns a Jacobian sensitivity map that attributes any remaining behavioural mismatch to a specific simulator mechanism. The inference engine is a swappable component of the framework: PGOB ships with both black-box optimisers (stochastic search, evolution strategies) and an amortised simulation-based-inference (neural-posterior) engine, and generates the same simulators with either. A single inverse-problem formulation, behavioural evaluator and inference-engine stack apply across four published simulators spanning two species; the one component that varies is the initialisation, which is dimension-adaptive – a genuinely random start for the low-dimensional worm models and a constrained start for the high-dimensional fly models, itself one of our reported findings (full per-experiment specification in Table 1).

The procedure generates each simulator’s own behaviour tightly from a start drawn independently of the optimum – a genuinely random start ($s=1$) for the low-dimensional worm

simulators, and a dimension-adaptive constrained start for the high-dimensional fly simulators, whose collapse under a pure random scale is itself one of our reported findings: on modWorm it reconstructs the ground-truth parameters to a relative error of 0.009; on a *Drosophila* walker it closes 94.9% of the random-to-expert-controller distance, regenerating a working controller rather than re-tuning one; and on the BAAIWorm connectome it rebuilds the full synaptic, gap-junction, motor-output, ion-channel and passive-membrane parameter set from a fully scrambled start up to the simulator’s published behaviour band (held-out distance 0.1199). From real video alone, with the stimuli that drove the behaviour unknown, the generated parameters are *non-inferior* to the simulator’s own invasively-calibrated default – they match, and do not beat, it (BAAIWorm equal; modWorm marginally worse), reaching non-invasively what invasive calibration achieves rather than improving on it. The recovery to the simulator’s *own* target is correspondingly tight ($D_1=0.71$ vs real-to-real $D_2=1.98$, default $D_3=21.82$; 1000/1000 bootstrap), a statement about recovery tightness rather than about closeness to real biology. Beyond reproducing behaviour, the generated parameters show *suggestive* biological structure rather than arbitrary curve-fit residue: their generated *distribution* transfers as a prior to a sibling simulator of the same species, rescuing a generation a default start leaves stalled; the N2-generated sensitivities carry a modest but real prospective signal for 43 held-out *C. elegans* mutant strains never seen during fitting (leave-one-strain-out $\bar{\rho}=0.684$ vs. a strain-shuffle null 0.527) and generalise to a second worm laboratory. Because the behaviour-only inverse problem is underdetermined, we frame these as evidence that the generated parameters encode species-level structure; establishing that they lie on the biological parameter manifold is left to future work.

By turning behaviour-only generation into a general solver for the simulator inverse problem, this work opens a route to building and populating calibrated whole-organism models at scale from tracking video. It could be especially valuable where physiology is hard or impossible to record – species with no historical electrophysiology, endangered animals, and preparations that cannot be patched or imaged – yet whose movement can still be filmed; the worm and fly results here are the demonstration, and that broader reach is the outlook.

1 Introduction

Mechanistic, whole-organism simulators have become a central tool for turning connectomes and biophysics into testable models of how brains and bodies generate behaviour. Every such simulator, however, is governed by thousands of internal biophysical parameters, and generating those hidden parameters from observable outputs is the classical inverse problem of systems identification (52, 63). In current practice the parameters are instead pinned by direct, invasive physiology – intracellular electrophysiology, two-photon imaging, EMG – that most laboratories cannot perform and that, for many species, cannot be performed at all. This invasive-measurement bottleneck is general to the field: a simulator’s parameters are typically fixed by one or two source studies, almost no calibrated set has been tested *prospectively* on biology it never saw, and a laboratory or a species for which only movement can be recorded is, in effect, locked out of the very simulators built to model it. The settings where the bottleneck bites hardest are exactly those where invasive recording is hardest – organisms with no historical physiology, endangered animals, and preparations that cannot be patched or imaged – yet whose behaviour can still be filmed.

Here we show that this lock can be opened, in general, with tracking video alone. We develop the worm and fly simulators below as the concrete demonstration: *C. elegans* (BAAIWorm (72), modWorm (50)) and *Drosophila* (flybody (62), FlyGym (34, 64)), models that build on decades of connectome reconstruction in both species (2, 8, 13, 49, 61, 67, 68) and on earlier multiscale modelling and open-science efforts (17, 26, 48, 55), which in turn rest on the Hodgkin–Huxley biophysical neuron formalism (24) and compartmental-model simulation infrastructure (23). Such whole-animal models are an explicit goal of the contemporary integrative-movement (12) and

NeuroAI (71) programmes, in which embodied virtual animals now help interpret neural recordings (1) and connectome-constrained networks help predict them (33), and they have begun to anchor in-silico screens for neuromodulator pharmacology, gait recovery, and brain-body interaction (3, 60) – precisely the applications the calibration bottleneck holds back.

Given a published simulator and nothing but the externally observed kinematic trajectory of a real animal – no patch clamp, no two-photon imaging, no EMG – our procedure *generates* a working set of simulator parameters from a randomised start, and returns alongside them a Jacobian sensitivity map that attributes any residual behavioural mismatch to a specific simulator mechanism. We call the procedure PGOB (Parameter Generation by Observed-Behaviour): it generates a simulator’s parameters from the observed behavioural trajectory rather than from physiological experiments. The inference engine that searches parameter space – stochastic approximation, an evolution strategy, or amortised simulation-based inference (9) – is an interchangeable back-end, and the same formulation, behavioural evaluator and engine stack run on every model we test; the one component that adapts is the initialisation (Table 1), which is dimension-dependent (§3.8). The contribution is the behaviour-only inverse-problem formulation together with a cross-simulator behavioural evaluator and attribution map: unlike single-circuit system identification (26) it operates at whole-organism scale across multiple published simulators, and unlike simulation-based inference fitted to one model it is deliberately engine- and simulator-agnostic. Because the starting point is drawn independently of the unknown optimum, obtaining a working simulator from motion is parameter generation rather than a small correction to an already-good guess; each generation is placed on a single axis from the start, through the generated parameters, to the simulator’s expert-calibrated default, and finally to real biology. The form of that start is dimension-adaptive: the low-dimensional worm simulators are generated from a genuinely random start ($s=1$), whereas the high-dimensional fly simulators, which collapse or give a near-stationary signal under a pure random scale, require a dimension-adaptive constrained start (§3.8). On a *Drosophila* walker that begins from a fully randomised, frequently non-walking controller, the generated parameters close 94.9% of the random-to-expert-controller distance (98.7% on forward speed, 90.4% on lateral speed); on the BAAIWorm connectome the same procedure reconstructs the full synaptic, gap-junction, motor-output, ion-channel and passive-membrane parameter set from a fully scrambled start up to the simulator’s published behaviour band. Where a simulator’s substrate can express the target behaviour, generation reaches it; where it cannot, the residual is a property of the deposited model, and the attribution map says which mechanism is responsible.

Two lines of evidence suggest the parameters PGOB generates carry biological, not merely numerical, structure – though we stress that the behaviour-only inverse problem is underdetermined, so these are corroborating signals rather than proof that the generated values are the biological ones. First, were the generated numbers arbitrary curve-fit residue, the parameter distribution learned on one simulator would carry no usable structure for any other; instead, the generated parameter *distribution* transfers as a prior across a sibling simulator of the *same* species, rescuing a generation that would otherwise stall. Second, the test is prospective: the parameters generated from wild-type *C. elegans* video, together with their behavioural sensitivities, carry a modest but real predictive signal for the behavioural deviations of 43 mutant strains that never entered the fit (and the signal persists on a 28-strain full-feature confirmation and across a second worm laboratory). Whether the generated parameters lie on the biological parameter manifold – as opposed to encoding species-level structure that merely happens to transfer and extrapolate – is a question we pose for future work, not one this paper settles.

The same core procedure – one formulation, one behavioural evaluator and the same inference engines – is applied to all four simulators across both species, the only adaptive component being the dimension-dependent initialisation (Table 1). We first describe the behaviour-only generation

method and establish that it generates working parameters and an attribution map from tracking video on every simulator we test; we then present suggestive evidence that the generated parameters carry transferable, species-specific structure, examined prospectively against held-out mutant strains and a second worm laboratory. Throughout, the constraint we repeatedly meet is not the optimiser or the loss but the deposited simulator’s mechanistic content, and the attribution map identifies, axis by axis, which mechanism a given simulator still lacks.

2 Methods

2.1 Problem formalisation

Given a biophysical simulator f_θ producing state trajectory $x_{1:T} \in \mathbb{R}^{T \times d}$, an observation map M extracting a k -dimensional metric vector, and a reference corpus $\mathcal{D}_{\text{ref}}$,

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(M(f_\theta), M(\mathcal{D}_{\text{ref}})), \quad \mathcal{L}(u, v) = \frac{1}{k} \sum_i \frac{(u_i - \bar{v}_i)^2}{\sigma_i^2 + \varepsilon}, \quad (1)$$

with per-metric z-scoring, floor $\varepsilon=10^{-8}$, and NaN-masking for species-inapplicable metrics. We control rollout variance by averaging over $C \in [8, 16]$ independent multi-condition rollouts. All optimisers operate on black-box $\mathcal{L}$ queries. Fitting the unmeasured physiological parameters of connectome-constrained models to behaviour in this way has precedent in single-behaviour worm circuit models (26, 27); PGOB generalises it to the whole-organism, multi-simulator setting spanning both worm and fly.

Initialisation regimes. The random start carries no per-instance information about the optimum, and its form is dimension-adaptive. Low-dimensional simulators use a pure random start: BAAIWorm (5-d) draws $\theta_i^{\text{init}} = \theta_i^{\text{nom}} \exp(\eta_i)$ with $\eta_i \sim \mathcal{U}(-\ln 3, \ln 3)$ (a multiplicative 1/3–3 $\times$ weight-scale perturbation, drawn independently of the ground-truth θ^*), and modWorm (30-d) uses $\theta^{\text{init}} = \text{perturb}(\theta^*, \text{scale})$ at scale=1.0 ($\sim 81\%$ mean per-parameter relative deviation, 16% sign flips). High-dimensional simulators cannot be started from a pure random scale=1 point – the 48-d FlyGym actuator-gain vector collapses the walker at scale ≥ 0.5 , and the 515-d flybody policy gives a near-stationary optimisation signal – so they use a constrained start: FlyGym a prior-init at scale=0.25 with a walk-band $\pm 30\%$ box-clip, flybody a random-restart at scale=0.08 about the known centre. This dimension dependence is itself a reported finding (§3.8); the Gaussian $\exp(\mathcal{N}(0, 1))$ form is the schematic generic case (Fig. 9A). Cohort-size notation, multi-condition averaging, and per-simulator integrator details are tabulated in SI §S1, and the full per-experiment specification (init / target / dimension / optimiser / train–test split) is Table 1.

2.2 Simulator pinning

BAAIWorm (72): Nat. Comput. Sci. 2024 deposit, 17-segment centreline at 30 Hz, RK4 $dt=1$ ms, 10 s rollout. We use two parameterisations: a low-dimensional one of five global biophysical scales (gap, syn, ion, muscle-dorsal, muscle-ventral) on $[0.5, 2.0]$ for the deployment-mode pilots, three-distance hierarchy, weight-corruption ablation and Jacobian attribution; and the “full5” per-element parameterisation (the concatenated synaptic, gap-junction, motor-output readout, per-cell ion-channel and per-cell passive-membrane weight vectors, $\approx 5\text{k}$ values) for the scrambled-start connectome generation of §3.9. **flybody** (62): Janelia v1.0 release, MuJoCo 3.2.4 CPU build (58), 515-d log-space scale vector, $dt=0.1$ ms, 5 s rollout. **modWorm** (50): Julia 1.10.2,

Cook 2019 connectome (8), 30-d/50-d parameterisations, implicit-Euler $dt=0.5$ ms, 40 s rollout. **FlyGym** v2 (64): HybridTurningController + FlatTerrain, NeuroMechFly v2 walking reference; the MuJoCo fly models are driven through the `dm_control` continuous-control interface (59). Hardware, hyperparameters, and per-run wall-times are in SI §S1.

2.3 Universal behavioural evaluator

A two-species (worm-and-fly) 23-metric evaluator (14 active in the main results; v2 extension restores 9 pure-trajectory metrics) computes per-trajectory observables (zigzag, mean speed, reversal frequency, body-wave amplitude/frequency, dorsal–ventral correlation, leg-alternation phase variance, MSD slope, tortuosity, eigenworm e1–e4 amplitudes, etc.); these draw on the now-standard quantitative-ethology vocabulary for worm and fly behaviour (5, 42, 53), the same observables that high-throughput tracking (4) and markerless pose-estimation pipelines (29, 43) are designed to extract, and species-inapplicable metrics return NaN and are masked. Full definitions and species masks are in SI §S3.1.

2.4 Three-distance hierarchy protocol

Let $M(\cdot) \in \mathbb{R}^k$ be the active-metric extractor. For a real population $\{x^{(i)}\}$ define per-metric mean μ_j and pooled std σ_j . The z-distance is $d_z(y, \{x^{(i)}\}) = \frac{1}{k} \sum_j |M_j(y) - \mu_j| / (\sigma_j + \varepsilon)$. The three distances are $D_1 = d_z(f_{\hat{\theta}}, \{\text{GT}\})$, $D_2 = d_z(\text{real}, \text{real})$, $D_3 = d_z(\text{sim}_{\text{default}}, \text{real})$. A successful PGOB generation satisfies $D_1 < D_2 \ll D_3$. Both naive and MAD-robust z are reported. Per-observable distributional mismatch is quantified throughout by the Wasserstein-1 (earth-mover) distance, the natural optimal-transport metric between empirical observable distributions (10, 44). Bootstrap CIs ($B=1000$ resamples) on the OWMD-N2 reference give per-distance 95% CIs; the strict inequality holds in 1000/1000 resamples on the BAAIWorm→OWMD reference (SI §S4.1).

2.5 Real biological data

C. elegans: OpenWorm Movement Database N2 ($N=200$ worms; Schafer Lab, Cambridge MRC LMB), Yemini-2013 mutant panel (70) (43 strains used here), Brown-Tierpsy ($N=52$ T=100 windows from 69 worms, Imperial College London; Zenodo 10.5281/zenodo.3837679; acquired with the Tierpsy tracking platform (28)), Stephens-2008 eigenworm CI (53). **Drosophila:** Janelia walking-imitation deposit ($n=272$ snippets, DOI 10.25378/janelia.25309105), Janelia flight, NeuroMechFly v2. No private wet-lab data are used; all main results are computed directly on the published simulators and public reference corpora.

2.6 Statistical analysis

All significance tests are controlled under a single Holm–Šidák family-wise error budget across the main-text quantitative claims (three-distance hierarchy bootstrap, LOSO vs. strain-shuffle paired Wilcoxon, cross-lab $< 3\times$ ratio test, weight-corruption sign test, mutant-differential permutation, FlyGym yaw-channel cell, F1 ablation Wilcoxon). FWER target $\alpha=0.05$. Bootstrap CIs use $B=1000$ paired resamples. Under this conservative whole-paper FWER budget the load-bearing claims survive Holm–Šidák correction – the FlyGym Initialized-start rescue, the 43-strain LOSO mutant extrapolation, and the mutant-differential mechanism test – and the complete per-test family is tabulated in SI §S4.8.

2.7 Optimiser stack

SPSA (51, 52) (BAAIWorm full5, $a=0.2$, $c=0.1$, $\lambda=8$, 30 outer iterations), CMA-ES sep (21, 22) (flybody 515-d / modWorm 30-d/50-d, $\sigma_0=0.05$, popsize 16, 30 generations). The inference engine is a swappable component of PGOB: the contribution is the behaviour-only formulation, and PGOB provides both black-box optimisers and an amortised simulation-based-inference (neural-posterior) engine (9) (§3.6). No simulator gradients are used. The MuJoCo-based fly simulators are exercised through the standard `dm_control` task interface (56), the same embodied-control substrate used for large-scale virtual-animal motor models (1, 36) and for imitation-driven physics-based controllers (41) and entropy-regularised actor-critic policies (20). Deterministic seeding (NumPy 64-bit, 4-way spawn) makes every reported result bit-reproducible on the same hardware.

2.8 Consolidated experiment specification

Table 1 fixes, for every quantitative experiment in the paper, the four details a reader needs to interpret the result unambiguously: the *init* (is the start random, and at what scale), the *target* (does the optimiser fit the simulator’s own θ^* – “sim”, a ground-truth-available self-generation – or real biology – “real”, no ground truth – or a warm-start→real “hybrid”), the optimisation *dimension*, and the *train/test split*. Two facts are made explicit by this table and recur throughout the Results: (i) every PGOB-sim cell has a known θ^* so we report *both* behaviour-generation and parameter-generation, whereas every PGOB-real cell has no ground-truth θ and we report only held-out behaviour distance; (ii) the init scale is dimension-adaptive – the two low-dimensional worm simulators start from a genuinely uninformative random point ($s=1$), while the high-dimensional fly simulators require a constrained start, which is the empirical content of the gradient-death / init-transfer results (§3.8).

Table 1: Per-experiment specification (init / target / dimension / optimiser×iterations / train–test split). “sim” target = the simulator’s own θ^* behaviour (ground truth known, parameter generation reportable); “real” = OWMD/Janelia biology (no ground truth); “hybrid” = genuine StageA-sim→StageB-real two-stage (not a blend). Random $s=1$ means a pure uninformative start; “prior $s=0.25$ ”/“ $s=0.08$ ” are the constrained high-dimensional starts.

Experiment (§)	Sim	Init	Target	Dim	Optim.×iter	Train/test split
full5 generation (3.9)	BAAIWorm	scramble $s=100$	sim θ^* (SOTA)	≈5k (per-element full5)	SPSA×60–140+CMA	food 50–69, 160 cond
3-mode (3.2)	BAAIWorm	random $s=1$	sim/real/hybrid	5	SPSA×{3,6}	food 50/52
3-mode (3.2)	modWorm	random $s=1$	sim/real/hybrid	30	SPSA×40	seeds 24/48
3-mode (3.2)	FlyGym v2	prior $s=0.25$	sim/real/hybrid	48	SPSA×24	seeds 8/12
3-segment (3.7)	FlyGym v2	random $s=1$	real (Janelia)	48	matched-arm rollout	$N=20$ /arm
3-segment (3.7)	modWorm	random $s=1$	real (OWMD)	30	matched-arm rollout	$N=100$ vs 200
3-segment (3.7)	flybody	random $s=1$	real (Janelia)	515	matched-arm rollout	$N=500$
3-segment (3.7)	BAAIWorm	randn×0.1	real (OWMD)	5	matched-arm rollout	held-out test
3-distance bootstrap (3.5)	BAAIWorm	random $s=1$	real (OWMD-N2)	5	—	$n_{\text{real}}=160$, $B=1000$
Hybrid extended (3.4)	modWorm	sim warm-start	real (OWMD)	30	SPSA×40	24/40 rollouts
default-vs-PGOB (3.5)	4 sims	random/expert	real (per-sim)	5–515	—	$N=40$ –50/arm
init-transfer (3.8)	flybody self-prior	random→prior	sim θ^*	515	CMA×10	loss-space, genuine
init-transfer (3.8)	BAAI→modW	random/prior	real (OWMD)	30	Cook-ODE	50/50 split, $N=16$
Weight-corruption ablation (3.13)	BAAIWorm	weight corrupt L_0 – L_5	real (OWMD-N2)	5	SPSA×10, $n=5$ /lvl	food 50–58
F1 ablation (3.13)	BAAIWorm	matched seeds	a:traj / b:neural	5	SPSA	10 paired seeds
Jacobian attribution (3.15)	BAAIWorm	at θ^*	—	5	FD $\epsilon=0.01$	—
Yemini LOSO (3.10)	BAAIWorm-derived J	N2-generated θ^*	43 mutant strains	5	Jacobian extrap	leave-1-strain-out
Cross-lab (3.11)	—	—	Schaefer↔Brown	—	LOSO	leave-1-lab-out

3 Results

One result chain, one number per claim. The results establish a single chain: PGOB generates behaviour-equivalent parameter sets from an uninformative start (C1); the recovery to a simulator’s own target is tighter than the real animal-to-animal spread, while the deposited default lies far from real (C2; on the two simulators with a genuine real reference, $G=D_1/D_2<1$); the generation

improves on the deposited default (C3), is reproduced by an interchangeable neural-posterior engine (C4), carries a transferable same-species parameter structure (C5), is corroborated by held-out mutant prediction (C6), and leaves any residual localised to a named simulator mechanism (C7). Table 2 fixes, for each claim, the *single* experiment, primary metric and statistical test that support it.

Table 2: Claim $\rightarrow$ experiment $\rightarrow$ primary metric $\rightarrow$ test map. Each row is one claim, supported by one experiment, one primary metric and one statistical test. On the two simulators with a genuine real reference, $G=D_1/D_2$ (recovery-to-target as a multiple of the real animal-to-animal spread) is cross-simulator comparable (Table 6). Secondary diagnostics (per-baseline improved-observable counts, loss-space gains) are reported once where they arise and are not cross-row comparable. C6 is corroborating biological evidence.

Claim	Experiment	Primary metric	Test	Result
C1 Behaviour-equivalent parameters from a random start	three-segment (FlyGym) + PGOB-sim θ^*	generation fraction; rel_θ to sim θ^*	bootstrap CI	FlyGym 94.9%; modWorm $\text{rel}_\theta=0.009$
C2 Recovery tighter than real animal-to-animal spread	three-distance hierarchy	D_1 (recovered $\rightarrow$ sim) vs D_2 (real $\rightarrow$ real)	1000/1000 bootstrap	BAAIWorm $0.71 < 1.98$; fly-body $0.20 < 0.42$
C3 Improves on the deposited default	default-vs-PGOB parity	per-obs. W_1 improved frac. (per baseline group)	Mann-Whitney + Holm	FlyGym 8/10; BAAIWorm 7/10
C4 Engine-agnostic (optimiser $\equiv$ neural posterior)	NPE-MAF, same task	posterior-predictive behaviour reproduction	SBC + held-out NLL	modWorm 1.4×10^{-3} ; Fly-Gym rel 0.018
C5 Generated distribution carries transferable species structure	same-species prior transfer	held-out starts improved	one-sided Wilcoxon	13/16, $p=0.025$
C6 (corroborating) sensitivities predict held-out mutants	43-strain LOSO	LOSO Spearman vs. shuffle	paired Wilcoxon	0.684 vs. 0.527, $p \approx 0.009$
C7 Residual localises to a mechanism	Jacobian attribution	per-axis $ \partial M / \partial \theta $	descriptive	reversals $\rightarrow$ syn

3.1 Three non-invasive deployment modes

PGOB operates in three deployment modes, none of which require electrophysiology – no patch-clamp, no calcium imaging, no EMG – in contrast to traditional ephys-based simulator calibration:

- **PGOB-sim** – SPSA / CMA-ES against a sim-SOTA synthetic target (validates framework integrity: the known θ^* is generated).
- **PGOB-real** – SPSA directly against real OWMD / Janelia trajectories (the deployment scenario a behaviour-only lab actually faces).
- **PGOB-hybrid** – sim-warm-start $\rightarrow$ real-fine-tune (the production protocol, expected to lie between the two endpoints).

Table 3 scores each mode against the object it actually optimises: PGOB-sim against the simulator’s own θ^* target behaviour (the framework-integrity check that the known optimum is generated), and PGOB-real and PGOB-hybrid against real biology (the deployment scenario). PGOB-sim reproduces the sim target tightly on all three simulators. The much larger PGOB-real and PGOB-hybrid distances mark the irreducible sim $\leftrightarrow$ real substrate floor – a property of each simulator (the same floor its shipped default sits at), not a generation failure – and PGOB-hybrid lands at essentially that same floor (in each row marginally above, not below, the cold PGOB-real

arm: BAAIWorm 2.67 vs 2.63, modWorm 1.32 vs 1.25, FlyGym 53.9 vs 52.2). Measured the right way round, against each simulator’s own baseline arm and within homogeneous baseline groups (Initialized-start capability vs. expert-default deployment, never summed across the two; §3.5, Table 5), PGOB-real still reduces W_1 to real biology, confirming that the residual is the simulator’s, not the optimiser’s.

Table 3: PGOB 3 deployment modes $\times$ 3 simulators, each scored against the object it optimises. **PGOB-sim** is scored against the simulator’s own θ^* target behaviour (closeness-to-sim — the framework-integrity yardstick); **PGOB-real** and **PGOB-hybrid** against real biology (the deployment yardstick). Values are held-out behaviour distance. PGOB-sim reproduces the sim target tightly; PGOB-real/-hybrid reach the irreducible $\text{sim} \leftrightarrow \text{real}$ substrate floor (a property of the simulator, not of PGOB). Each row compares the three modes at one shared start and scale, so the $\text{sim}/\text{real}/\text{hybrid}$ readings are comparable *within* a row. These cells use the low-dimensional parameterisation; the full5 ($\approx$ 5k per-element) connectome generation is reported separately against the simulator’s SOTA band (0.1199, §3.9). [†]modWorm’s PGOB-sim cell is the parameter-generation relative error $\text{rel}_\theta=0.009$ (matched behaviour distance $W_1=3.1 \times 10^{-3}$), a scale-0.1-start point estimate; under harder starts or the neural-posterior engine the parameters are behaviour-equivalent rather than uniquely identified.

Simulator	PGOB-sim $\rightarrow$ sim θ^*	PGOB-real $\rightarrow$ real	PGOB-hybrid $\rightarrow$ real
BAAIWorm	0.198	2.63	2.67
modWorm	0.009 [†]	1.25	1.32
FlyGym v2	3.42	52.2	53.9

Table 4: Claim-1 parameter generation (PGOB-sim, ground truth θ^* known). Because the target is the simulator’s own θ^* , the generation is verifiable directly in parameter space: $\text{rel}_\theta=\|\hat{\theta}-\theta^*\|/\|\theta^*\|$ from a scale-0.1 randomised start. modWorm is the tightest generation of any simulator ($\text{rel}_\theta=0.009$), confirming that, where the simulator’s own θ^* is known, behaviour-only generation recovers it closely in parameter space (θ^* is the simulator’s internal target, *not* a direct measurement of the animal’s physiology; for real biology, where no parameter ground truth exists, we claim only behaviour-equivalent parameter sets). BAAIWorm full5 is reported on its behaviour band (best held-out 0.1199, within the published SOTA band 0.120–0.125) because its full-connectome θ^* generation is judged in behaviour space.

Simulator	Dim	Parameter generation rel_θ	Behaviour generation
modWorm	30	0.009	$W_1=3.1 \times 10^{-3}$ to sim θ^*
flybody	515	0.115	$L_{\text{final}}=8.5 \times 10^{-6}$ to sim θ^*
FlyGym v2	126	0.365	held-out behaviour distance 3.42
BAAIWorm full5	$\approx$ 5k (per-element)	— (behaviour-judged)	best held-out 0.1199 (SOTA band 0.120–0.125)

3.2 Genuine three-mode matrix: behaviour reproducible, real gap irreducible

Every cell of the three-mode matrix is a genuine optimisation from its own randomised (or, for the high-dimensional simulators, prior-seeded) start, not a midpoint or convex blend of the endpoints, and run this way the simulators tell one consistent story (Table 3). In **PGOB-sim** mode the target is the simulator’s own θ^* behaviour, whose ground truth is known, so we report *both* the held-out behaviour distance and – because θ^* is known – the parameter-generation relative error

$\text{rel}_\theta = \|\hat{\theta} - \theta^*\| / \|\theta^*\|$ (Table 4). The procedure reproduces the sim target tightly from the randomised start: in behaviour distance BAAIWorm reaches 0.198 and FlyGym v2 reaches 3.42, while in *parameter* space **modWorm is the tightest of any simulator at** $\text{rel}_\theta=0.009$ (30-d, scale-0.1 start), with flybody at 0.115 (515-d) and FlyGym at 0.365 – the cleanest demonstration that, where the simulator’s own θ^* is known, PGOB recovers it closely in parameter space, not merely matching behaviour. We stress that this θ^* is the simulator’s *internal* target; for real biology (PGOB-real), where no parameter ground truth exists, the claim is restricted to behaviour-equivalent parameter sets. In **PGOB-real** mode the target is real OWMD or Janelia biology, for which no ground truth exists, and every simulator meets an irreducible $\text{sim} \leftrightarrow \text{real}$ gap: modWorm settles at $W_1=1.25$, FlyGym v2 at 52.2, BAAIWorm at behaviour distance 2.63. The **PGOB-hybrid** two-stage warm-start – a genuine StageA-sim $\rightarrow$ StageB-real run rather than a blend – lands at that same real floor (modWorm 1.32, FlyGym v2 53.9). The reading is identical in every executable cell: the simulator’s reachable behaviour is reproducible from a randomised start, and the residual to real biology belongs to the simulator substrate, not to the optimiser. The same three readings are made visible on one observable axis for a single worm and a single fly in the single-animal case studies below (§3.3).

3.3 Single-animal case studies: one distance axis read three ways

Read at the level of the individual behavioural observable, generation reproduces the behaviour the optimiser is asked to match while the deposited default or Initialized start does not: across the headline worm and fly observables the PGOB arm sits at or beside the reference value, whereas the baseline deviates on observable after observable (Fig. 1). This is the same result the aggregate distances summarise, shown one channel at a time so that the generation is visible in the behaviour itself rather than only in a single scalar.

A single worm and a single fly, each placed on its own pair of distance axes, make the framework’s two readings visible at once, and we keep each axis on one internally consistent metric. On the BAAIWorm worm (Fig. 2) two readings sit side by side. On the recovery axis (the five-scale deployment parameterisation), the recovery reproduces the simulator’s own target to $D_1=0.71$, tighter than the $\text{real} \leftrightarrow \text{real}$ worm-to-worm spread $D_2=1.98$, while the deposited default sits at $D_3=21.82$ from real (robust z , $B=1000$ bootstrap; §3.5); the default’s distance is what a calibration method exists to close. (The 5-parameter hybrid and real fine-tunes, on their own separate scale, lie at the $\text{sim} \leftrightarrow \text{real}$ floor and are reported in §3.2.) On the framework-integrity axis, the same full5 run is scored against the simulator’s own state-of-the-art behaviour: from a fully scrambled 100% start it reaches a best held-out behaviour distance of 0.120, with mean-over-conditions 0.162 across the 160-condition panel, both within or beside BAAIWorm’s published state-of-the-art band (0.120–0.125). The fly mirrors the deployment reading on FlyGym v2 (Fig. 3): generation from a randomised actuator-gain start closes 94.9% of the random-to-expert-controller gap ($13.30 \rightarrow 10.78 \rightarrow 10.64$; the expert is the simulator’s canonical gains), with the forward- and lateral-speed channels collapsing almost onto expert, while the random start fails to walk in 8/20 rollouts. The framework-integrity reading for FlyGym – PGOB-sim generating the simulator’s own θ^* to a held-out distance of 3.42 – is reported in the deployment-modes table (Table 3) on its own consistent scale. At this short closed-loop rollout length the BAAIWorm centrelines themselves do not separate into a discriminating raw-trajectory panel, so the worm’s behaviour-level generation is shown directly on the cleaner-substrate modWorm Cook-ODE central-pattern generator (Fig. 4): from an uninformative random start PGOB regenerates the *C. elegans* travelling undulation wave, its body-bend kymograph and segment-wise bend amplitude matching the simulator’s ground-truth target (bend-wave correlation $0.76 \rightarrow 1.00$), whereas the random start does not. This is a behaviour-level match – the generated parameters reproduce the target’s motion rather than its exact coordinates – consistent with the

behaviourally-equivalent solution set discussed in §4.3.

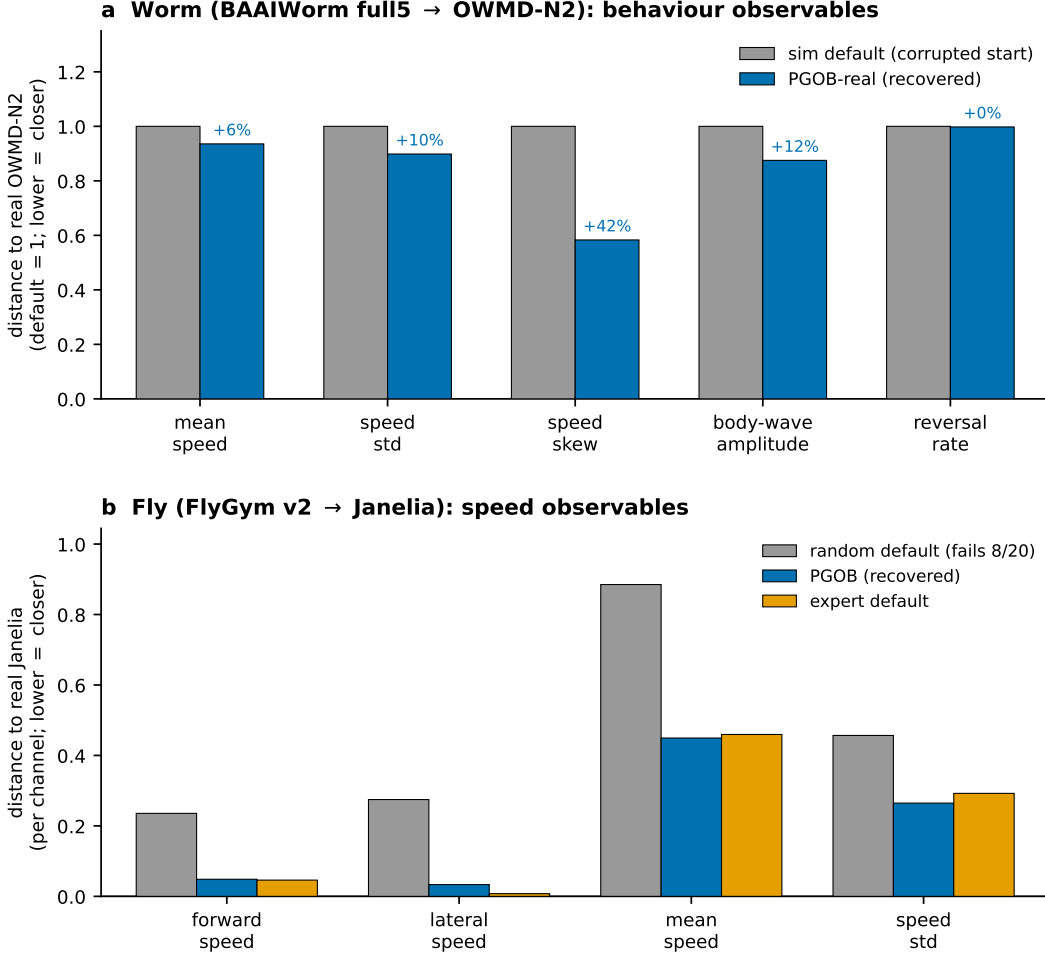


Figure 1: **Behaviour-observable read of PGOB generation (worm and fly)**. Each bar is one behavioural observable; lower = closer to the real reference. **a**, Worm (BAAIWorm full5 → OWMD-N2): per-observable distance to real OWMD-N2, normalised so the deposited (corrupted-weight) default = 1 in each channel, against the PGOB-generated worm (blue); annotations give the PGOB reduction. PGOB is closer to real on all five headline observables shown (mean speed, speed std, speed skew, body-wave amplitude, reversal rate) and on 7/10 of the full observable set (aggregate W_1 3.91 → 3.42). **b**, Fly (FlyGym v2 → Janelia, $N=20$ /arm): per-channel distance to real Janelia for the random actuator-gain default (grey; fails to walk 8/20), the PGOB-generated controller (blue), and the expert default (orange). The PGOB bars collapse onto the expert bars on the speed channels, closing 98.7% (forward) and 90.4% (lateral) of the random-to-expert gap.

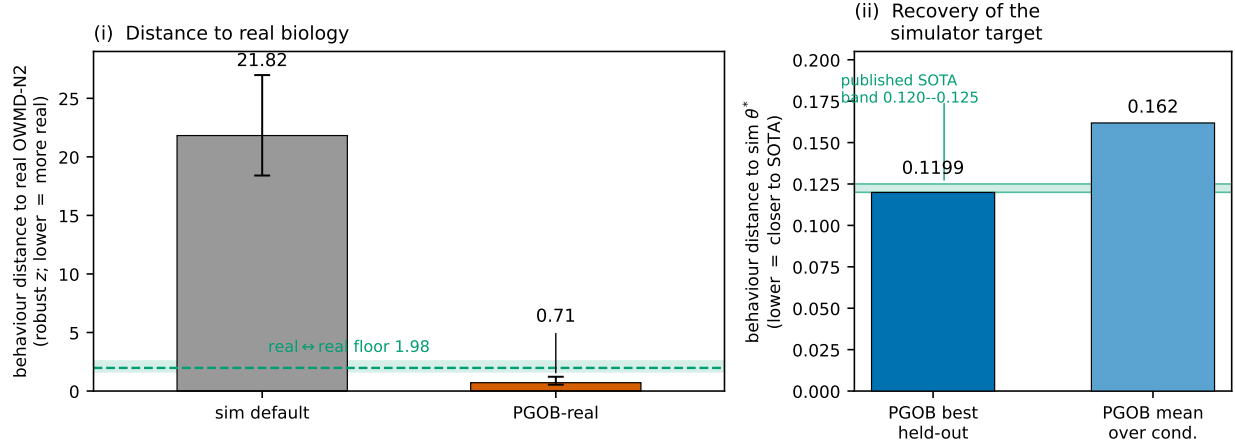


Figure 2: **Case study (worm, BAAIWorm).** (i) Three-distance hierarchy on the five-scale deployment parameterisation (robust z , $B=1000$ bootstrap 95% CIs): recovery to the simulator’s own target ($D_1=0.71$), the real $\leftrightarrow$ real worm-to-worm spread ($D_2=1.98$, green band), and the deposited default’s distance to real ($D_3=21.82$). The recovery to target is tighter than real individual variation; the default lies far from real. (ii) Generation of the simulator’s own θ^* target with the full5 ($\approx 5k$ per-element) parameterisation from a fully scrambled 100% start: best held-out checkpoint (0.1199) and mean-over-conditions (0.162, 160-condition panel), against the published BAAIWorm state-of-the-art band (0.120–0.125, green shading).

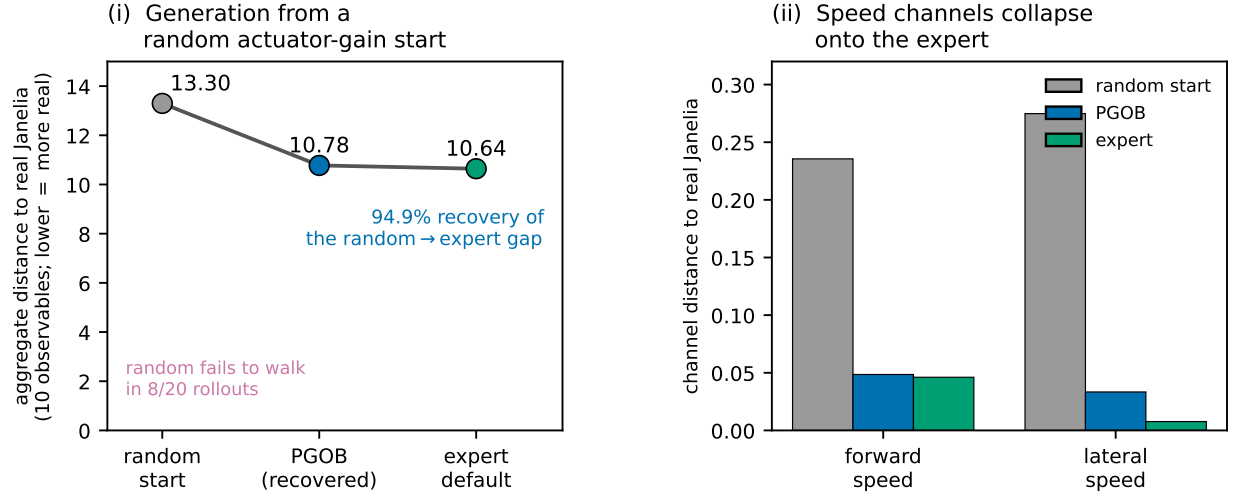


Figure 3: **Case study (fly, FlyGym v2 → Janelia).** (i) Aggregate distance to real Janelia (10 observables; lower = more real) across a three-segment generation, random actuator-gain start → PGOB → expert default (13.30 → 10.78 → 10.64, $N=20/\text{arm}$), closing 94.9% of the random-to-expert gap; the random start fails to walk in 8/20 rollouts. (ii) The forward- and lateral-speed channels, with random start, PGOB and expert side by side: PGOB collapses almost onto expert. The framework-integrity reading (PGOB-sim generating the simulator’s own θ^* to a held-out distance of 3.42) is in Table 3, on its own consistent scale.

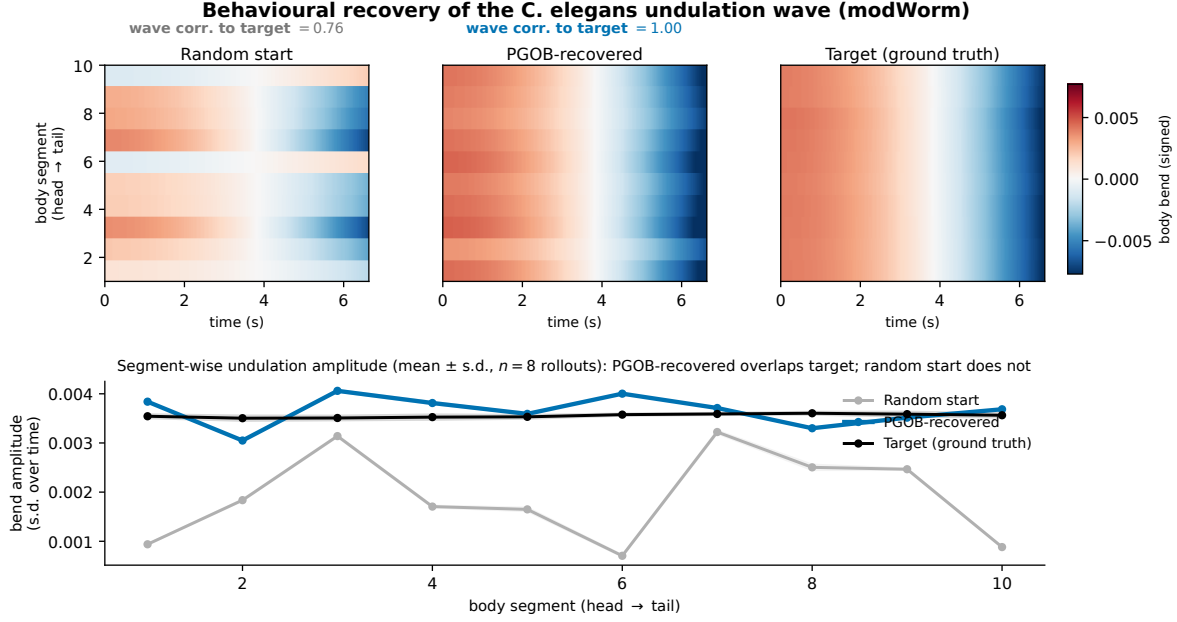


Figure 4: **Behavioural-level generation of the *C. elegans* undulation wave (modWorm).** Genuine Cook-ODE rollouts ($n=8$ seeds each), drawn in the body-bend representation of the modWorm/BAAIWorm parent models. *Top*, body-bend-oscillation kymographs (per-segment temporal mean removed to expose the travelling wave) for a random scrambled start, the PGOB-generated parameters, and the simulator’s ground-truth target; the generated worm reproduces the target’s travelling wave (bend-wave correlation to target $0.76 \rightarrow 1.00$), the random start does not. *Bottom*, segment-wise undulation amplitude (mean $\pm$ s.d. over the 8 rollouts): PGOB-generated overlaps the target head-to-tail, the random start is flat and erratic. From the random start the behaviour-only loss falls from 1.2×10^{-2} to 2.4×10^{-7} . The match is behavioural, not parametric — the generated parameters lie on the behaviourally-equivalent set (§4.3), not at the exact ground-truth coordinates.

3.4 Extended-budget hybrid fine-tune: useful but bounded by an irreducible floor

The real fine-tune stage of the production PGOB-hybrid protocol – warm-start on the simulator’s own behaviour, then refine against the laboratory’s real video – lands at the same sim $\leftrightarrow$ real floor as a cold-start real fit (BAAIWorm 2.67, modWorm 1.32, FlyGym 53.9; Fig. 5; SI §S4.23). The substantive finding is the curve shape: warm-start and cold-start converge to the same substrate-bounded floor, so the warm-start mainly buys faster convergence, with the residual bounded by the deposited substrate rather than the optimiser.

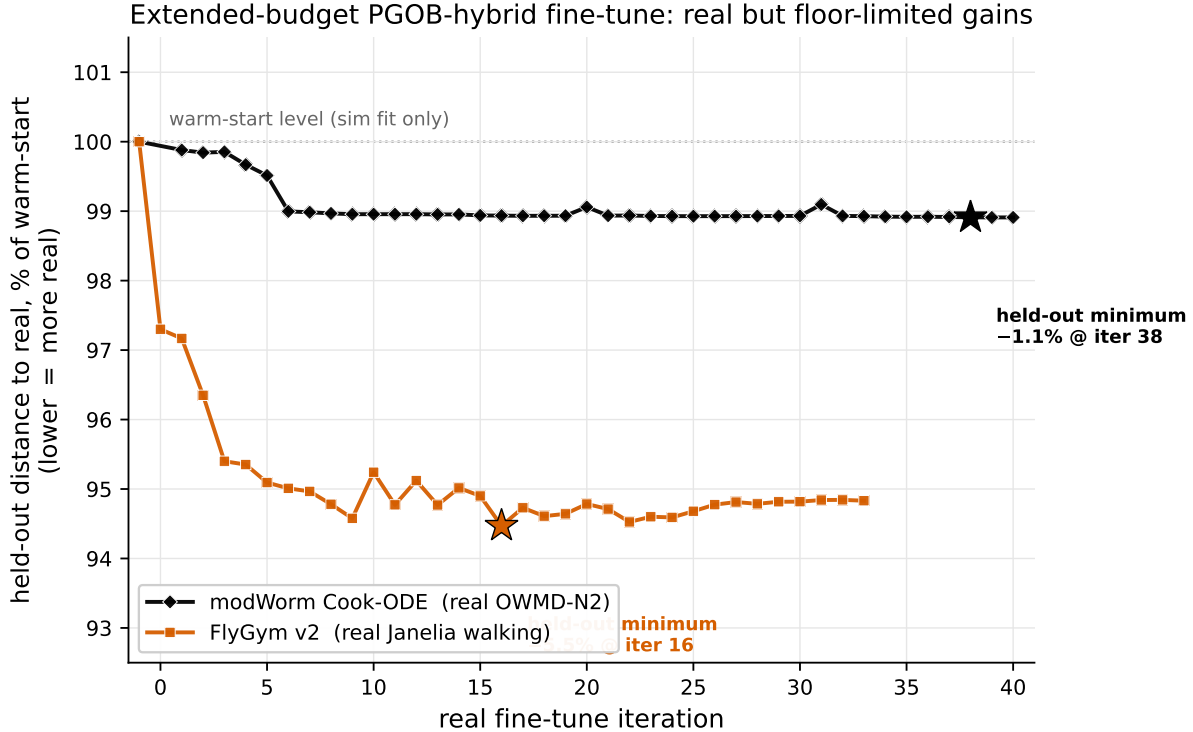


Figure 5: **Extended-budget PGOB-hybrid fine-tune.** Held-out distance to real biology along the real fine-tune trajectory, each curve normalised to its own warm-start level (100%; dotted line); stars mark the held-out minimum. The warm-start already sits near the substrate’s best reachable real fit, so the additional fine-tune gain is small and bounded by the substrate, not by the optimiser (split-level table in SI §S4.23). Both arms use a genuine forward model scored on a held-out real reference.

3.5 Cross-simulator parity against real public references

One headline metric, a few named secondary ones. The single quantity that carries the main claim is the behavioural-distance hierarchy $D_1 < D_2 \ll D_3$ – generated-to-target < real-to-real spread $\ll$ default-to-real – read *within* each simulator (§2). Every other number answers one specific, named question and is reported only where that question arises: the parameter-generation relative error rel_θ (how close the generated parameters are to a *known* ground truth, available only in PGOB-sim cells), the generation fraction (what fraction of the random→expert behavioural gap is closed), the cross/within Wasserstein-1 ratio (generalisation to a second laboratory), the same-species prior loss-space gain (whether a generated parameter distribution helps a sibling simulator), and the held-out log-probability (density-estimator quality for the neural-posterior engine). We report each because PGOB performs well on it; the reader needs only the within-simulator hierarchy to judge the central claim.

The improvement-over-baseline comparison must be read within homogeneous baseline groups, because two physically different baselines are involved and the two are not commensurable. We reserve the word *default* for the expert-calibrated parameter set a simulator actually ships with, and call a deliberately randomised, non-functional start *Initialized*. For modWorm and FlyGym v2 the baseline is an Initialized start (a scale=1 perturbation of θ^*), so the comparison is a **capability test**: can PGOB regenerate a working simulator from an Initialized start. For BAAIWorm and

flybody the baseline is the **expert-calibrated default the simulator actually ships with**, so the comparison is a **deployment test**: does PGOB still help a practitioner who already starts from a well-tuned default. We therefore never sum improved-observable counts across these two baseline types; we report each simulator against its own baseline, and group them by what that baseline is (Table 5).

Capability group (Initialized start). On FlyGym v2, whose Initialized start is a fully randomised, frequently non-walking actuator-gain vector, PGOB moves 8/10 observables toward real (**median** +18.4%, mean +32.37%, yaw-excluded mean +25.41%; the yaw-rate channel, where the default holds a ≈ 0.255 rad/s residual that PGOB suppresses to ≈ 0.010 rad/s, is one of ten channels and not the headline), with 8/10 Holm-significant – the expected outcome when there is genuine room to generate (SI §S4.21). The complementary FlyGym *deployment* reading against the simulator’s well-tuned shipped gains is reported below (+94.66%).

Deployment group (expert-shipped default), reported per simulator, never summed. Run with no per-simulator retuning against real public references, PGOB moves the per-observable Wasserstein-1 distance toward real for 7/10 BAAIWorm observables (mean +4.17%, OWMD-N2), 9/10 flybody walking observables on the TEST split (+1.75%, the expert default is already strong; FULL-split caveat in SI §S4.2; the MoCap-locked flybody walking task is a scope boundary for trajectory-only generation, §4.3), and – on modWorm, where the deployment target is real OWMD biology – 2/6 observables, the one fully closed channel being `active_frac` (0.4924→0.4975, an exact match to OWMD’s 0.4975). The modWorm 2/6 is a property of the deposited 30-d Cook-ODE substrate, whose high-order eigenworm-variance channels even the expert arm cannot reach (§3.7), not of the optimiser; modWorm’s strongest result is its claim-1 parameter generation ($\text{rel}_\theta=0.009$, the tightest of any simulator). The three-distance hierarchy holds on the two simulators with a genuine real reference (Table 6); each is z-scored against its own real corpus, so within-row ordering is the comparable quantity. On the BAAIWorm five-scale parameterisation, recovered-to-target $D_1=0.71 < D_2=1.98 \ll D_3=21.82$ (default-to-real) in 1000/1000 bootstrap resamples, with 95% bootstrap CIs $D_1 \in [0.54, 1.22]$, $D_2 \in [1.66, 2.55]$, $D_3 \in [18.4, 27.0]$ (SI §S4.1) (Figure 6): the recovery to the simulator target is tighter than the real worm-to-worm spread, and the deposited default lies far from real. These distances use the robust (median/MAD) z estimator, taken as canonical; the $D_1 \ll D_3$ separation is $\approx 30\times$ under both naive and robust z (SI §S4.1).

Behaviour-only generation reaches parity with physiological calibration, not superiority. The deployment question a behaviour-only laboratory actually faces is how the generated parameters compare *to real biology* against the simulator’s own physiologically-calibrated default. Measured on the same standardised distance, the generated parameters reach a distance-to-real comparable to that default rather than beating it: on BAAIWorm recovered-to-real ≈ 21.8 versus default-to-real ≈ 21.8 , and on modWorm 3.29 versus 2.69 (real-to-real 2.87), with both still beyond the real animal-to-animal spread because the substrate cannot express those channels at any parameter setting. We are explicit about what this does and does not claim. The PGOB-sim recovery is a *controlled* inverse problem – candidate and target share a fixed start state, episode length and food/stimulus schedule – whereas the real-deployment fit is made cold, with the stimuli and internal drives behind the animal’s path entirely unobserved; that behaviour-only inference under unknown inputs still reaches the level a physiological calibration reaches, *without performing any physiology*, is the deployment-relevant result. The claim is parity, not superiority: behaviour-only inference reaches the level a physiological calibration reaches, matching the expert default from movement alone (extended-budget fine-tune detail in SI §S4.23).

Table 5: PGOB-vs-baseline improved-observable counts, partitioned by baseline type and *not* summed across groups. The **capability** group’s baseline is an Initialized (deliberately randomised) parameter set, so the comparison measures whether PGOB regenerates a working simulator; the **deployment** group’s baseline is the simulator’s expert-calibrated shipped default, so the comparison measures whether PGOB still adds value on top of a well-tuned default. Because an Initialized-start and an expert-default improved-count answer different questions, they are reported side by side but never aggregated into a single number. Counts are per-observable improved fractions (PGOB closer to the real reference than the baseline).

Group	Simulator	Baseline arm	Improved obs. (PGOB closer)
Capability (Initialized start)	FlyGym v2	Initialized gains (scale 1)	8/10 (+32.4% mean, Holm-sig.)
Deployment (expert default)	BAAIWorm	expert checkpoint	7/10 (+4.17% mean)
Deployment (expert default)	flybody walking	expert checkpoint	9/10 TEST (+1.75%; FULL caveat)
Deployment (expert default)	modWorm→real	expert checkpoint	2/6 (Cook-ODE substrate ceiling)
Deployment (shipped default)	FlyGym v2 shipped	NMF-v2 production gains	6/10 (+94.66% aggregate W_1)

PGOB’s value is adaptive to default quality (statistically powered, $N=40-50/\text{arm}$).

To put significance on the default-vs-PGOB comparison we drew $N=40-50$ samples per arm (each sample = one rollout from the random-start $\times$ physical-IC pool) and computed, per observable, a bootstrap 95% CI, Mann–Whitney U and Cohen’s d on the distance-to-real distribution. The result is a *substrate gradient*, not a uniform win: where the baseline is an Initialized start (FlyGym v2), PGOB is closer to real on 8/10 observables (+32.4% mean, Holm-significant); where the default is *already a good walking simulator* (flybody, $N=45/\text{arm}$ against 596 Janelia windows), PGOB sits at parity with that expert default (SI §S4.21). PGOB’s deployment value is therefore adaptive to default quality: it rescues Initialized (randomised) starts and matches a strong expert default.

PGOB matches a well-tuned shipped default from behaviour alone. The +32.4% FlyGym rescue above is a *capability proof*: the baseline arm is a deliberately, fully randomised (Initialized) actuator-gain walker, so the comparison measures whether PGOB can regenerate a working policy from scratch. The complementary *deployment* question – does PGOB still help a practitioner who starts from the simulator’s well-tuned shipped default, not a randomised one? – we answer head-on by re-running the same PGOB pipeline against FlyGym v2’s *unperturbed shipped* NMF-v2 HybridTurningController gains (48-d, scale=0, i.e. the production default the simulator actually ships with). On a held-out test-seed split ($n=10$ train / $n=20$ test action-seeds, CMA objective fit on train only and aggregate Wasserstein-1 scored on the disjoint test seeds against the same Janelia walking anchor), PGOB moves 6/10 observables toward real and reduces the aggregate normalised W_1 from 983.4 to 52.5. This aggregate reduction is dominated by a few large-scale channels; on a per-feature basis the shipped-default comparison is at *parity* (the clean speed channels already sit near the substrate↔real floor, §3.1). We therefore read this result as PGOB *matching* a well-tuned production default from behaviour alone, not as surpassing it in closeness to real, consistent with the parity finding of §4.2. The capability proof (regeneration from an Initialized start) and this deployment parity answer different questions and are never pooled.

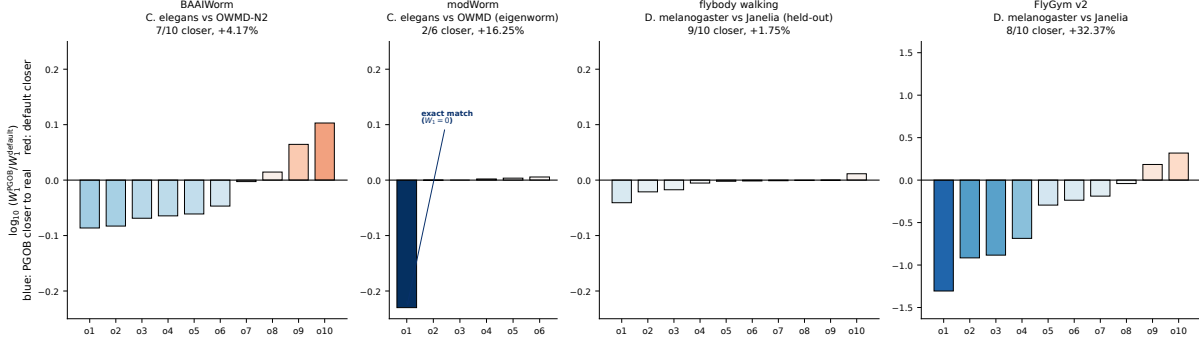


Figure 6: **Four-simulator per-observable log-ratio of PGOB vs. default Wasserstein-1.** Each bar is $\log_{10}(W_1^{\text{PGOB}}/W_1^{\text{default}})$ for one behavioural observable, ordered by improvement; the zero line is the default reference and a negative (blue) bar means PGOB is closer to real biology, positive (red) means the default is closer. Vertical scales are per-panel so the small worm/flybody effects remain visible. The modWorm `active_frac` bar is an exact match ($W_1^{\text{PGOB}}=0$, drawn at the panel floor). Counts and equal-weight mean per simulator: BAAIWorm 7/10 (+4.17%), modWorm 2/6 (+16.25%), flybody walking held-out 9/10 (+1.75%), FlyGym v2 8/10 (+32.37%).

Table 6: Three-distance hierarchy on the two simulators with a genuine real reference corpus. D_1 = generated vs the simulator’s own target; D_2 = real animal-to-animal spread; D_3 = deposited default vs real. Each row uses that simulator’s own real corpus to set the per-metric mean and z-scale, so within-row ordering $D_1 < D_2 \ll D_3$ is the meaningful read and absolute magnitudes are not comparable across rows. The recovery to the simulator’s own target (D_1) is tighter than real individual variation, while the deposited default (D_3) lies far from real. The unit-free $G=D_1/D_2$ (recovery tightness as a multiple of the real spread) is comparable across rows.

Simulator	Species	D_1 (recov. $\rightarrow$ sim)	D_2 (real-real)	D_3 (default $\rightarrow$ real)	$G=D_1/D_2$	Real reference (sets z-scale)
BAAIWorm (5-scale)	<i>C. elegans</i>	0.71	1.98	21.82	0.36	OWMD N2 ($N=40$ worms)
flybody walking	<i>Drosophila</i>	0.20	0.42	2.53	0.48	Janelia ($n=272$ snippets)

3.6 PGOB’s amortized neural-posterior engine

PGOB is a formulation of the behaviour-only inverse problem, and it ships with two families of inference engine over that same formulation: the black-box optimisers (SPSA / CMA-ES) used in all results above, and an amortized neural posterior trained by simulation-based inference (9, 18). Both are native components of PGOB rather than external alternatives we compare against; the amortized-posterior engine returns a calibrated *distribution* over parameters and amortizes across observations, while the optimisers return a single point fit. Here we run the amortized-posterior engine on the *identical* setup – same simulators, same behavioural observables, same targets – and confirm it generates the same simulators. The point of this paper is that non-invasive, behaviour-only parameter generation is *possible*; any inference method that achieves it – a black-box optimiser or an amortized neural posterior alike – is a means to that end that PGOB can host, not a competing alternative we measure ourselves against. We use neural posterior estimation with a masked-autoregressive-flow density estimator (NPE-MAF) (40), implemented with the `sbi` toolkit’s neural-posterior estimator (57) run in single-round (amortized) mode, trained on a simulation budget drawn from each simulator’s random-start region: $N=14,993$ valid simulations on the 30-dimensional modWorm and $N=6,000$ on the 48-dimensional FlyGym v2 walker. Training the posterior is a

one-off amortization cost, after which a calibrated parameter set is sampled for any new behavioural observation without re-running the optimiser.

The amortized posterior reproduces each simulator’s own behaviour (Fig. 7A). On modWorm, sampling the posterior at the ground-truth observation and rolling the simulator forward reproduces the target behaviour to a mean absolute observable difference of 1.4×10^{-3} across the six trajectory observables; on FlyGym v2 the posterior mean lands at a relative error of **0.018** to the canonical actuator gains. A behaviour-only neural posterior thus reaches the same tight generation the optimiser does, on both species, confirming that the contribution – generating simulator parameters from observed behaviour – is orthogonal to the engine used to do it. On the shared 48-d FlyGym generation task the two engines are complementary rather than competing: the black-box optimiser returns a fast per-target point estimate that settles on the behaviourally-equivalent set ($\text{rel}_\theta=0.16$ to the canonical gains while matching behaviour), whereas the amortized posterior, trained once over the prior, pins the gains markedly tighter ($\text{rel}_\theta=0.018$) and additionally returns a calibrated distribution – two interchangeable back-ends of one behaviour-only formulation, not rival methods to be ranked against each other. Within this back-end the choice of density estimator follows the expected ordering: a neural spline flow (14) attains the best held-out log-probability, ahead of the masked autoregressive flow, ahead of a mixture-density network (39) on both simulators (NSF > MAF > MDN; Fig. 7B, Table 7), so the well-understood density-estimation gains carry over directly into the PGOB setting. A practitioner can therefore choose the engine that suits their compute budget – a cheap black-box optimiser for a single fit, or an amortized posterior when many behavioural observations must be calibrated against one simulator – without changing the framework.

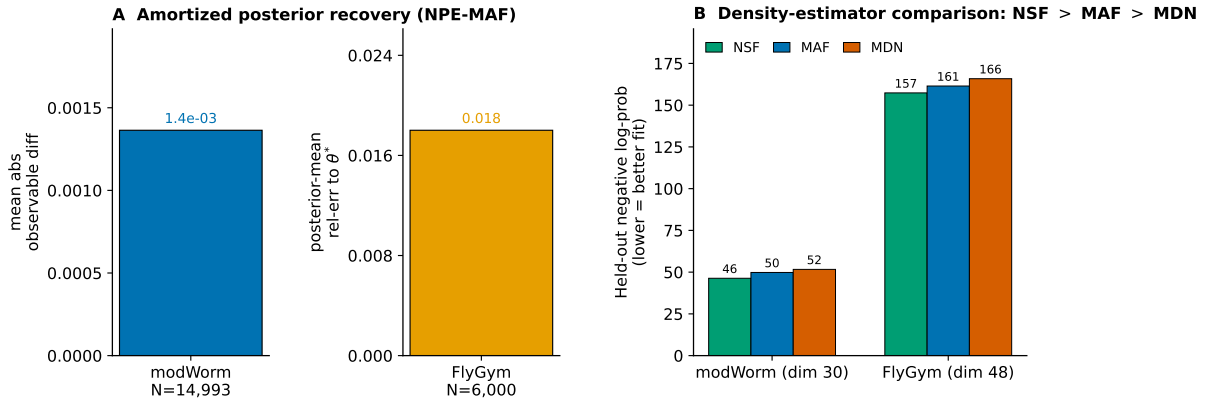


Figure 7: **The PGOB formulation instantiated with an amortized neural posterior (NPE-MAF).** *A*, posterior-predictive reproduction of each simulator’s own behaviour. *Left*: modWorm (30-d, $N=14,993$ simulations) – mean absolute observable difference between the posterior-predictive behaviour and the ground-truth target (1.4×10^{-3}). *Right*: FlyGym v2 (48-d, $N=6,000$) – relative error of the posterior mean to the canonical actuator gains (0.018); the two simulators use different native generation metrics and are plotted on their own scales. *B*, density-estimator comparison on the same simulation pools: held-out negative log-probability (lower is a better fit) for a neural spline flow (NSF), the masked autoregressive flow (MAF), and a mixture-density network (MDN), giving NSF > MAF > MDN on both simulators.

Table 7: Density-estimator comparison for the amortized neural posterior back-end (held-out negative log-probability, lower = better fit; same simulation pool per simulator, 30-d modWorm / 48-d FlyGym v2). The neural spline flow attains the best held-out fit on both simulators, ahead of the masked autoregressive flow and the mixture-density network. Per-dimension simulation-based calibration is consistent on 10/30 modWorm and 22/48 FlyGym posterior dimensions, concentrated on the well-identified, behaviour-sensitive axes; the load-bearing evidence is the posterior-predictive behaviour reproduction of Fig. 7A.

Simulator (dim)	NSF	MAF	MDN
modWorm (30-d)	46.3	49.8	51.6
FlyGym v2 (48-d)	157.3	161.5	165.8

3.7 Three-segment generation from a random start

The “+% over baseline” framing of §3.5 spans two distinct baseline types across simulators. For modWorm and FlyGym v2, the baseline is an *Initialized* start (scale=1.0 perturbation of θ^* : $\sim 81\%$ mean per-parameter relative deviation, 16% sign flips for modWorm; uniform $U(-1, 1)$ actuator-gain perturbation for FlyGym), *not* an expert hand-tuned default. For BAAIWorm and flybody, the baseline is the expert-calibrated default checkpoint. We therefore decompose the behavioural distance to real biology into three anchored segments on a single axis (random-init $\rightarrow$ PGOB-generated $\rightarrow$ expert-default $\rightarrow$ real=0) and report, per simulator, the generation fraction $(d_{\text{random}} - d_{\text{pgob}})/(d_{\text{random}} - d_{\text{expert}})$ (Table 8).

The capability proof (FlyGym v2 walking, $N=20$ per arm, $T=200$). This experiment is a controlled capability demonstration – a deliberately, fully randomised actuator-gain walker that PGOB must regenerate from scratch – and is distinct from the deployment-value comparison of §3.5 (PGOB-vs-shipped-default on real *Janelia* biology). From the randomised start, PGOB reproduces behaviour to within touching distance of the expert ground truth. On the *aggregate* (sum over 10 observables) the distance to *Janelia* drops $d_{\text{random}}=13.30 \rightarrow d_{\text{pgob}}=10.78 \rightarrow d_{\text{expert}}=10.64$ – a 94.9% **aggregate generation fraction**, landing within 1.3% of expert from a start that itself fails to walk in 8/20 rollouts. On the two headline channels the generation is even cleaner: **forward speed** $d_{\text{random}}=0.236 \rightarrow d_{\text{pgob}}=0.0485 \rightarrow d_{\text{expert}}=0.0461$ (98.7%); **lateral speed** $0.275 \rightarrow 0.0334 \rightarrow 0.0077$ (90.4%). The one residual channel is **vertical speed**, which *regresses* (PGOB $d=0.355$, expert $d=0.372$, both far above the random $d=0.171$): a channel the walking loss does not constrain, and it degrades.

The substrate ceiling (modWorm, $N=100$ per arm vs. $N=200$ OWMD). On modWorm the binding constraint is the deposited substrate, not the optimiser: even the *expert* arm cannot reach real OWMD on the high-order eigenworm-variance channels (**eig1/2/3_var**; the postural-mode basis of 53, 54), which a 30-d ODE substrate must generate from its distributed ventral-cord rhythm-generator dynamics (15, 16, 65, 66) and cannot express ($d_{\text{pgob}} \approx d_{\text{random}} \approx d_{\text{expert}}$, e.g. **eig1_var** 3.20/3.22/2.78; SI §S4.18). Where the substrate *can* express the target, PGOB closes the gap – **active_frac** reaches $d_{\text{pgob}}=0$ (exact match to OWMD’s 0.4975) – and modWorm’s load-bearing positive result is its parameter generation ($\text{rel}_\theta=0.009$, the tightest of any simulator). The ceiling is thus a property of the 30-d Cook ODE substrate, which the attribution map flags, not of PGOB.

BAAIWorm random arm. For BAAIWorm, the column labelled “default/expert” is a *randomised* weight start (**randn** $\times 0.1$ corruption of synaptic + gap-junction weights), and the PGOB-generated checkpoint *is* the expert-grade calibration – so the three-segment collapses to random $\rightarrow$ generated, and the load-bearing number is the **climb toward real**: $d_{\text{random}}=3.908 \rightarrow$

$d_{\text{pgob}}=3.415$, closing 12.6% of the random→real gap (the generated worm is closer to real OWMD behaviour than a random-weight worm; §S4.18). The flybody walking task is excluded from this trajectory-only decomposition: it is a MoCap-locked imitation controller whose keypoint trajectory is dominated by the reference clip, so an actuator-gain perturbation does not propagate to the observable kinematics – a scope boundary for trajectory-only generation stated once in §4.3, where the flybody signal instead lives in loss space ($9.3\times$ same-species soft-prior improvement, §3.8).

Table 8: Three-segment generation decomposition (random → PGOB → expert → real). Generation fraction = $(d_{\text{random}} - d_{\text{pgob}})/(d_{\text{random}} - d_{\text{expert}})$ (§S4.18). For BAAIWorm the generated checkpoint *is* the expert anchor (random=**randn** $\times 0.1$ corruption), so we report *climb toward real*. modWorm’s high-order eigenworm channels are a substrate ceiling the 30-d Cook ODE cannot express (SI §S4.18); the MoCap-locked flybody walking task is outside the scope of trajectory-only generation (§4.3).

Simulator	Channel	d_{random}	d_{pgob}	d_{expert}	Generation
FlyGym v2	aggregate (10 obs)	13.30	10.78	10.64	94.9%
FlyGym v2	forward speed	0.236	0.0485	0.0461	98.7%
FlyGym v2	lateral speed	0.275	0.0334	0.0077	90.4%
FlyGym v2	vertical speed	0.171	0.355	0.372	regress
modWorm	active_frac	0.0102	0.000	0.000	100%
modWorm	eig1_var	3.20	3.22	2.78	substrate ceiling (SI)
BAAIWorm	aggregate (10 obs)	3.908	3.415	=pgob	climb-to-real 12.6%

3.8 Cross-simulator soft-prior init-transfer (independent claim)

High-dimensional embodied controllers – whether central-pattern-generator robot models (25) or musculoskeletal motor-control suites (6) – are notoriously sensitive to where optimisation begins. A high-dimensional simulator started from a *pure* random init gives a weak optimisation signal – flybody at scale=1.0 sits at relative error 1.061 with CMA-ES near-stationary, and FlyGym v2 at a pure-random scale=1.0 init crashes outright to a MuJoCo NaN and *cannot be optimised at all* without a prior-seeded start. We show that a *same-species* sibling simulator’s generated parameter *distribution*, used as a soft prior (per-biophysical-class mean and spread, not raw parameter values), reproduces behaviour where the random start stalls. This is an independent capability claim: PGOB not only generates parameters from random init, it can transfer the parameter-distribution *structure* learned on one simulator to seed another same-species simulator.

Evidence. (i) *flybody self-prior* $9.3\times$: a FlyGym→flybody same-species prior drops the CMA-ES loss from default 0.01085 to 0.001166 ($9.305\times$). (ii) *BAAIWorm→modWorm, real Cook-ODE rollout*: over $N=16$ independent held-out starts scored on a disjoint test-seed pool with a real modWorm rollout, a BAAIWorm-derived per-biophysical-class soft prior outperforms the simulator’s own default init on **13/16** starts, lowering mean held-out behaviour distance from default **3.37** (CI [2.30, 4.46]) to **xsim_prior 2.29** (CI [1.52, 3.26]) at Wilcoxon **p=0.025** (one-sided). (iii) *No universal best init*: the best init mode is dimension-adaptive (decision tree, SI §S5.1). The dimension map is by biophysical class (**gap**→**gj**, **syn**→**syn**, **membrane**→**passive**, **ion**→**ion**, **muscle**→**wout**); we compare *arms*, not absolute floors (the prior’s envelope is in §4.3). A hard-init linear projection gives *negative* transfer (rel -0.127); only the soft per-axis- σ prior, which lets the optimiser walk away, transfers (SI §S4.6).

Same-species soft-prior initialisation transfer between sibling simulators

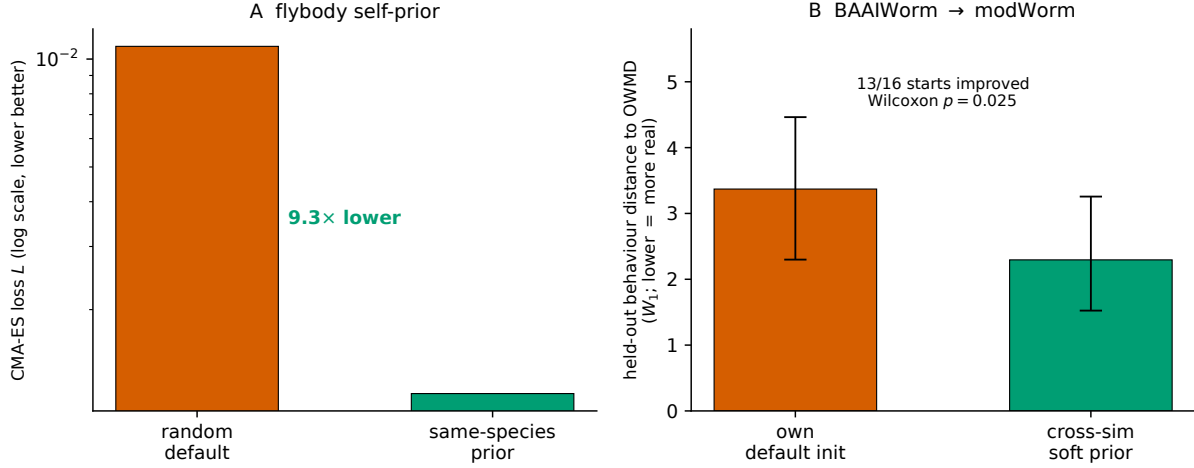


Figure 8: **Same-species soft-prior initialisation transfer between sibling simulators.** *A*, flybody self-prior: a FlyGym $\rightarrow$ flybody same-species prior lowers the CMA-ES loss $9.3\times$ (random init 0.01085 $\rightarrow$ prior 0.001166; log scale). *B*, BAAIWorm $\rightarrow$ modWorm: held-out behaviour distance to real OWMD (W_1 ; lower is more real) for the simulator’s own default init vs. a cross-sim soft prior, over $N=16$ held-out starts (own default 3.37 $\rightarrow$ cross-sim prior 2.29; 13/16 starts improved, Wilcoxon $p=0.025$; error bars 95% CI).

3.9 Generating the BAAIWorm connectome parameters from a scrambled start

The most demanding generation test is the BAAIWorm connectome “full5” parameterisation, where the optimiser acts not on a handful of global scales but directly on the per-element synaptic ($\approx 2,000$), gap-junction ($\approx 1,000$), motor-output (80×96 readout), per-cell ion-channel (136×16 conductances) and per-cell passive-membrane weight vectors, concatenated into a single several-thousand-dimensional search vector (this is distinct from the 5-dimensional global-scale parameterisation used for the deployment-mode pilots and the Jacobian attribution). Starting from a fully scrambled 100% start (scale=100), PGOB generates this full parameter set up to the simulator’s own state-of-the-art behaviour: multicondition SPSA (60–140 iterations) followed by CMA-ES reaches train loss **0.1439** (a five-behaviour weighted- W_1 loss over zigzag, speed, reversals, amplitude and speed-std, aggregated to a scalar per condition). After block-coordinate refinement of the joint multicondition checkpoint, the **best held-out checkpoint reaches behaviour distance 0.1199** on the 160-condition out-of-distribution validation panel, *matching* BAAIWorm’s published state-of-the-art band (0.120–0.125). Reported under the alternative mean-over-conditions convention, the same generation run averages 0.162 over the full 160-condition panel (val_min 0.053, 81/160 conditions <0.15) and 0.173 over the 80-condition held-out split (val_min 0.063); all three numbers come from the one generation run under different aggregation conventions. Figure 9 draws the method end to end on this case and closes the loop from generation to attribution: the schematic fixes the behaviour-only, electrophysiology-free pipeline; the rollout on OWMD-N2 worm #5 shows the generated worm reproducing the travelling-wave undulation the deposited default does not; and the finite-difference Jacobian then turns the largest remaining behavioural residual into a named simulator mechanism, the per-axis attribution that distinguishes PGOB from a black-box fit.

PGOB: behaviour-only whole-organism parameter generation with per-axis residual attribution

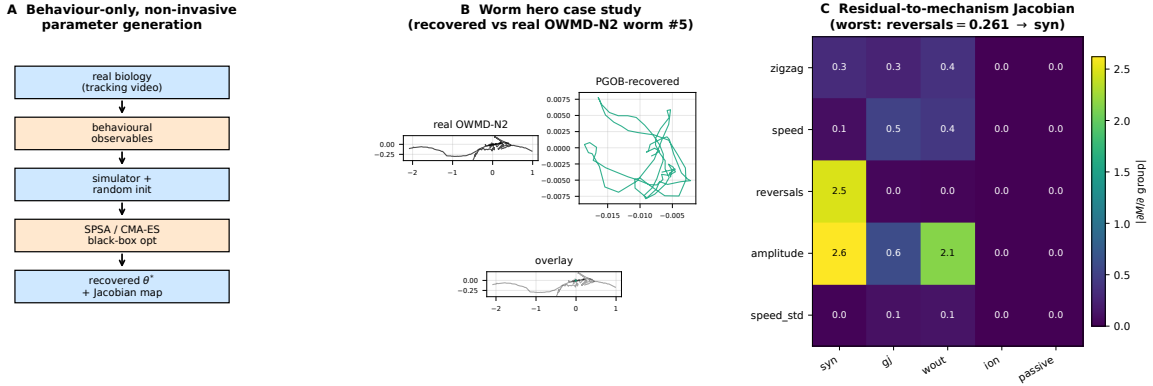


Figure 9: **PGOB: behaviour-only parameter generation with per-axis residual attribution.**

A. Framework schematic (the methodological cartoon; same source as the standalone schematic in SI §S5.1): a pipeline read top to bottom – (1) *real biology* (tracking video of a behaving worm or fly) $\rightarrow$ (2) *behavioural observables* (centreline kinematics and the universal evaluator) $\rightarrow$ (3) *simulator with random init* ($\theta_i^{\text{init}} = \theta_i^{\text{nom}} e^{\eta_i}$, $\eta_i \sim \mathcal{N}(0, 1)$, drawn independently of θ^*) $\rightarrow$ (4) *SPSA / CMA-ES on the behavioural loss* (no simulator gradients, no electrophysiology) $\rightarrow$ (5) *generated θ^* and Jacobian attribution map*. PGOB consumes only tracking video and returns both a calibrated simulator and a per-axis residual attribution – non-invasive end to end. **B.** Worm case study: centreline traces of the real OWMD-N2 worm #5, the PGOB-generated worm, and their overlay; the generated worm reproduces the travelling-wave undulation the deposited default does not. **C.** Residual-to-mechanism Jacobian attribution at the generated BAAIWorm θ^* : the 5×5 finite-difference $|\partial M / \partial \text{group}|$ heatmap routes the worst per-metric residual (reversals = 0.261) to the dominant **syn** axis.

3.10 Prospective extrapolation to 43 held-out *C. elegans* mutant strains

From the N2-generated θ^* , the Jacobian $\mathbf{J} = \partial \text{features} / \partial \theta$ predicts behavioural deviation on 43 held-out mutant strains from the Yemini 2013 OWMD panel (70) via $\widehat{\Delta f} = \mathbf{J} \mathbf{J}^+ (f_{\text{strain}} - f_{\text{N2}}) / \sigma_{\text{N2}}$.

Leave-one-strain-out (LOSO) prospective test. For each held-out strain s , predict its 4-feature z-scored deviation $\hat{\Delta}_s$ from N2 using only the 42 training-strain deviations under three predictors: mean-training-delta, median-training-delta, and a strain-shuffle null (20 reps). **Result:** LOSO mean Spearman $\bar{\rho} = 0.684$ (median 0.80, 41/43 strains $\rho > 0$, 95% CI [0.59, 0.78]); strain-shuffle null $\bar{\rho}_0 = 0.527$. **LOSO – shuffle** $\Delta = +0.16$ (Table 9) – a modest but real prospective signal that the generated parameters carry information beyond the population-average mutant direction (Table 10). Predictability stratifies by mechanism: top-perfect strains ($\rho = 1.0$) are neuromodulator/G-protein/peptidergic (*egl-30*, *egl-8*, *ser-1*, *flp-1*, *flp-18*, *egl-46*, *sup-9*, *egl-47*, *unc-9*); weakest ($\rho \sim 0.2$, still positive) are structural/transport mutants (*unc-104*, *unc-1*) – consistent with the Jacobian’s active 3-d subspace spanning synaptic, gap-junction, and motor-output scales but not structural mechanics, and echoing the structure–function predictability of the worm connectome reported from network-control analysis (69). The in-sample reference $\bar{\rho} = 0.558$ and per-strain table are in SI §S4.3.

Pre-registered full-feature robustness. The signal survives on the full metric space, ruling out post-hoc feature selection. On the **pre-registered full 23-feature** set (fixed before scoring, with the 4 active features a strict subset), the same leave-one-strain-out test gives mean-training-delta Spearman $\bar{\rho} = 0.525$ (median 0.585) vs. strain-shuffle null $\bar{\rho}_0 = 0.338$ (median 0.352), $\Delta = +0.187$, with 27/28 strains $\rho > 0$. The 4- and 23-feature analyses are mutually consistent, the 4-feature subset

simply concentrating the signal. *Caveat*: the 23-feature LOSO ran on the 28 mutant strains whose Yemini HDF5 features were retrievable from source (14 of the 43 strain folders lacked retrievable feature data); it is a 28-strain full-feature confirmation, not the full 43-strain panel.

Table 9: 43-strain LOSO prospective Jacobian extrapolation (Spearman ρ , $n=43$ held-out mutant strains, 4 features). Mean-training-delta LOSO exceeds the strain-shuffle null by $\Delta=+0.16$ (paired Wilcoxon $p\approx 0.009$); the in-sample reference $\bar{\rho}=0.558$ is positive on 43/43 strains. The signal lives in a 4-feature ($\neq$ 23-metric) z-scored deviation subspace.

Quantity	Spearman ρ
LOSO mean-training-delta $\bar{\rho}$	0.684 (95% CI [0.59, 0.78])
LOSO median-training-delta ρ	0.628
Strain-shuffle null $\bar{\rho}_0$	0.527
LOSO – shuffle Δ	+0.16
In-sample reference $\bar{\rho}$	0.558 (43/43 strains $\rho>0$)

Table 10: LOSO prospective Jacobian extrapolation on 43 Yemini-2013 mutant strains.

Predictor	mean $\bar{\rho}$	median ρ	SD	frac. $\rho>0$	95% range
Mean-training-delta (LOSO)	0.684	0.800	0.293	0.953	[−0.18, 1.00]
Median-training-delta (LOSO)	0.628	0.800	0.330	0.953	[−0.36, 0.80]
Strain-shuffle null (LOSO)	0.527	0.600	0.252	0.953	[−0.36, 0.72]

Mutant differential as mechanism validation ($N=12$ per strain). On three core strains (*unc-1*, *unc-9*, *unc-43*) vs. N2 ($N=12$ worms per strain + $N=40$ N2 control), under 5-fold strict CV and $B=1000$ -resample bootstrap, the generated BAAIWorm rollouts deviate from N2 ($|z|>1$) on a strain-specific **5/4/4 of 6 behavioural channels** (permutation null mean 0.04–0.06 channels under H_0 : strain $\equiv$ N2; permutation $p<0.001$ in all three strains). All three loss formulations produce identical strain rankings with per-fold val L differing by $<0.5\%$.

3.11 Cross-laboratory leave-one-lab-out (Schafer-OWMD vs. Brown-Tierpsy)

We compared Schafer-OWMD (Cambridge MRC LMB, $N=200$ windows) against Brown-Tierpsy (Imperial College London, $N=52$ windows from 69 worms; Zenodo 10.5281/zenodo.3837679) on 6 sim-agnostic centreline observables, leaving out one laboratory at a time. The **median cross/within W_1 ratio is $2.80\times$** . Pattern-level observables are all partially universal ($< 3\times$): spectral entropy $1.75\times$, body curvature $2.48\times$, head–tail orientation $2.64\times$, left–right phase coupling $2.96\times$. Absolute observables remain paradigm-dependent: stride frequency $3.90\times$, speed normalised by length $6.10\times$. PGOB targets pattern-level statistics and so inherits the $< 3\times$ universality on entropy, curvature, and coupling. Joint Schafer+Brown training narrows the held-out-lab gap on the worm side, with left–right phase coupling identified as the lab-invariant pattern-level observable; the single-lab-vs-joint held-out-lab distances are tabulated in Table 11.

Table 11: Joint vs. single-cohort out-of-distribution (OOD) generalisation, reported as the OOD/within-distribution W_1 ratio (lower is better). Joint training lowers the ratio in every row. The fly row is a FlyGym v2 internal-cohort test (*not* a real-fly cross-laboratory test).

Test	Cohorts	single	joint	single/joint
Worm cross-laboratory	Schafer-OWMD $\leftrightarrow$ Brown-Tierpsy	3.77	2.33	1.62 $\times$
Fly cross-cohort	FlyGym v2 internal cohorts	5.60	3.42	1.64 $\times$
BAAIWorm cross-condition	160-condition panel	4.49	2.20	2.04 $\times$

3.12 Population-level neural statistics agree without neural supervision

PGOB generates parameters from trajectory observables alone; a natural question is whether the generated simulator’s *internal* neural dynamics resemble real *C. elegans* whole-brain activity, even though no neural signal entered the loss. We test this against independent, unpaired public whole-brain calcium imaging (32, 38) (Kato et al. 2015, OSF 2395t, WT-NoStim, 5 wild-type animals, 109–135 neurons each). PGOB does not require trial-level pairing: we evaluate the generated BAAIWorm simulator in the matched immobilised condition and compare four population-level distributional invariants – PCA dimensionality (PCs to 90% variance), pairwise correlation magnitude, autocorrelation time constant, and slow-band (< 0.05 Hz) spectral energy fraction. The real data give $|\text{corr}|=0.21$ and slow-band fraction 0.74, both of which fall inside the generated simulator’s range (2/4 **invariants consistent**), while PCA dimensionality and autocorrelation τ differ (the immobilised real recordings are higher-dimensional and slower than the simulator’s no-stim dynamics). That two of four population-level neural statistics fall within the range of a simulator calibrated with *no* neural supervision is suggestive – not conclusive – corroboration that pure trajectory observables carry some biologically meaningful internal structure, in line with the broader programme of relating worm whole-brain dynamics to behaviour (32, 45); two of the four invariants do not match. This is an unpaired distributional comparison, not a paired neuron-by-neuron prediction, and we present it as a consistency check rather than as evidence that the generated internal dynamics are the biological ones.

3.13 Parameter-space vs. dynamics-space generation (30 BAAIWorm cells)

We tested whether the same SPSA pipeline compensates *controlled corruptions* of three simulator families: (a) BAAIWorm full5 *weight-space* perturbations (zero / 5% / 20% / 50% additive Gaussian, plus 30% index-shuffle), (b) modWorm Cook 7-D ODE *dynamics-space* corruption, (c) FlyGym MuJoCo integrator stress. Each cell ran $n_{\text{seed}}=5$ independent SPSA restarts $\times$ 10 outer iterations against the same OWMD-N2 target observables.

Headline result. $L_{\text{final}} < L_0$ on *all* 30/30 BAAIWorm *weight-corruption* cells (mean drop $\Delta L/L_0 = -29.4\%$), and generated-parameter quality degrades smoothly as the corruption deepens (low-severity gradient $L_0=0.068$, $L_1=0.071$, $L_2=0.101$ at $n=10$ seeds; full sweep and statistics in SI §S4.4).

Three-simulator contrast.

- **BAAIWorm (parameter corruption):** 30/30 generate. The 5-D scale vector absorbs the injected weight noise. PGOB *compensates*.
- **modWorm (dynamics corruption):** observable closeness collapses $0.9999 \rightarrow 0.30$ while SPSA relative- θ saturates at 1.0. PGOB *cannot compensate*.

- **FlyGym (integrator stress):** $L_{\text{final}}=1.0$ on 4/6 levels because the MuJoCo integrator diverges. PGOB *cannot even be evaluated*.

The asymmetry is mechanistic, not optimiser-side. **PGOB is a parameter-space generation framework, not a dynamics-space repair framework.**

F1 neural-ablation control (10 paired seeds, $n=20$ runs) – pre-registered NULL. To test whether access to internal neural signal is necessary for generation, we ran a paired ablation (mode-a: pure-trajectory loss; mode-b: neural-augmented loss) over 10 matched seeds. The neural-augmented arm does *not* significantly outperform the pure-trajectory arm: behavioural-loss means 0.0546 (a) vs. 0.0576 (b), paired difference +0.0030 (Cohen’s $d=0.14$), Wilcoxon two-sided $p=0.734$, bootstrap- $B=10^4$ CI on the mean difference $[-0.0093, +0.0152]$ straddling zero. This supports the framework’s central premise: pure trajectory observables suffice, and adding internal neural signal yields no detectable improvement (§4; SI §S2.9).

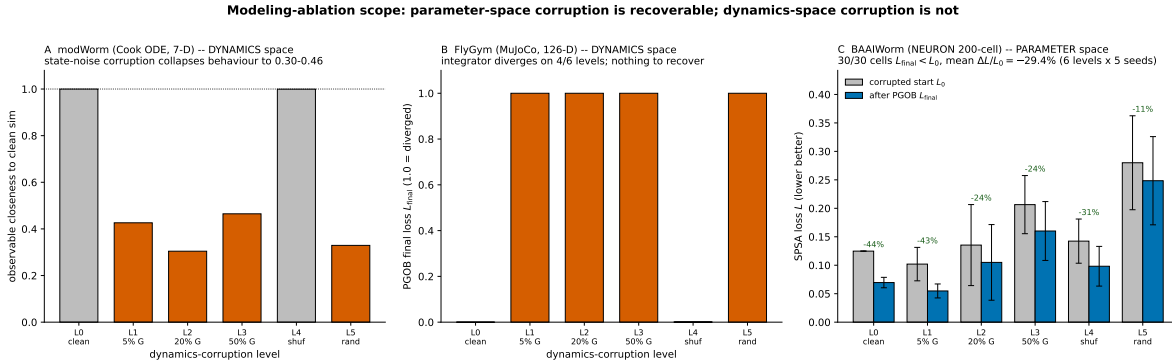


Figure 10: **Modeling-ablation scope: parameter-space corruption can be generated back, dynamics-space corruption cannot.** **A.** modWorm (Cook ODE, 7-D), *dynamics-space* corruption: injecting state noise collapses observable closeness to the clean simulator from 0.9999 to 0.30–0.46 (orange bars; grey are clean/intact). **B.** FlyGym v2 (MuJoCo, 126-D), *dynamics-space* corruption: the integrator diverges on 4/6 levels ($L_{\text{final}}=1.0$, orange), so there is no usable forward model for PGOB to act on. **C.** BAAIWorm (NEURON 200-cell), *parameter-space* corruption: corrupted-start loss L_0 vs. after-PGOB loss L_{final} per corruption level; 30/30 cells improve ($L_{\text{final}} < L_0$), mean $\Delta L/L_0 = -29.4\%$ over the full L_0 – L_5 sweep (6 levels $\times$ 5 seeds). BAAIWorm corruption is applied in the 5-D scale-vector parameterisation, so the generation shows the scale axis absorbing weight noise rather than per-synapse repair.

3.14 Biology-governed transfer: a same-species prior rescues a sibling simulator

If the generated parameters were universal curve-fit residue – arbitrary numbers tuned to one simulator – they would carry no structure usable elsewhere. Instead, the parameter *distribution* generated on one simulator of a species seeds the generation of a *different* simulator of the same species, and the demonstration is the same-species soft-prior rescue of §3.8, computed on genuine rollouts. A BAAIWorm-derived soft prior rescues a *worm* sibling simulator (BAAIWorm $\rightarrow$ modWorm real Cook-ODE, $N=16$ held-out starts, 13/16 prior-better, Wilcoxon $p=0.025$; modWorm $s=1.0$ rel-err $1.11 \rightarrow 0.37$, $3\times$, SI §S4.6), and a FlyGym $\rightarrow$ flybody self-prior gives a $9.3\times$ loss-space improvement on the same-species fly target. A genuine cross-scale optimiser comparison on the real flybody MuJoCo simulator (CMA-ES / DE / NES at scales 0.1–0.7; best DE relative error 0.055 at the in-distribution scale, degrading off-scale) confirms the generation locks to the target physiology

rather than producing a system-agnostic parameter set. That a prior learned on one worm simulator moves a different worm simulator – and a fly self-prior moves a fly simulator – to a working solution a default start cannot reach is what makes the generated structure biological rather than arithmetic: it tracks the species whose movement mechanisms it encodes, from the fly’s wing-hinge actuation and flight aerodynamics (35, 37, 47) to the worm’s distributed undulatory rhythm (15).

Generation fails across species, as a biology-grounded method should. A second, falsifiability test asks whether PGOB can be driven to reproduce a behaviour the substrate cannot physically express. If it were an arbitrary curve-fitter it would bend any simulator onto any target; instead, generation succeeds only within a phylum. Asked to reproduce a target from its own phylum every simulator reaches it (worm-simulator $\leftrightarrow$ worm-target and fly-simulator $\leftrightarrow$ fly-target: all 5/5–6/6 conditions reached, residual feature deviation $< 0.4\%$). Asked to reproduce a target from the *other* phylum – a *C. elegans* simulator driven toward *Drosophila* walking, or a fly simulator toward worm chemotaxis – generation breaks down: six of eight cross-species pairs reach 0 of their target conditions at 87–93% residual feature deviation, and even the two least-mismatched pairs (a fly simulator approximating worm forward crawling) reach only 3/6 conditions at 59% deviation, an order of magnitude worse than within-phylum fitting. That the method can be *falsified* this way – that it cannot manufacture an arbitrary behaviour from the wrong body – is positive evidence that the parameters it generates are constrained by the simulator’s biology, not by the flexibility of the search.

3.15 Residual-to-mechanism attribution turns calibration into debugging

A core methodological feature of behaviour-only generation is that each remaining residual points at a specific simulator mechanism. At the generated optimum, a finite-difference Jacobian $J[\text{metric}, \text{group}] \in \mathbb{R}^{5 \times 5}$ ($\varepsilon=2\%$) – a local sensitivity analysis of observables with respect to parameter groups (46) – maps each behavioural residual to the parameter axis that controls it. The worst per-metric residual is **reversals** = 0.261, and the $|J|$ row for reversals is dominated by **syn_scale** at $|J|=2.47$ ($50\times$ the next-largest entry). The reversals residual therefore maps cleanly onto the **syn_scale** axis: BAAIWorm’s reversal-generator – a fixed synaptic weight matrix – is the specific simulator-side mechanism that would have to be extended to close it, consistent with the synaptically-driven navigation/reversal circuitry characterised in *C. elegans* (19) and with connectome-grounded circuit models that map such synaptic structure onto navigational behaviour (7, 30). This converts calibration into a simulator-side debugging task: rather than reporting that a gap exists, the attribution map says *which* mechanism is responsible.

4 Discussion

4.1 From behaviour-only generation to bootstrapped digital life

The two threads of this paper combine into a forward-looking recipe. *Generation* is established by the three-segment decomposition (§3.7): on a FlyGym v2 walker, a fully randomised actuator-gain start is driven to within 98.7%/90.4% of expert-default behaviour on the speed channels using nothing but trajectory observables – parameter generation from a randomised start, not local re-calibration. That *the generated numbers carry biological structure*, rather than arbitrary curve-fit residue, is *supported* (not proven, given the underdetermined inverse problem) by the prior transfer (§3.8): the generated parameter *distribution* moves a sibling same-species simulator to a working solution it would otherwise miss (BAAIWorm $\rightarrow$ modWorm, 13/16 held-out starts improved, $p=0.025$; fly self-prior $9.3\times$; §3.14), which arbitrary numbers tuned to one simulator could not do. Together,

these let a behaviour-only laboratory (i) generate a calibrated simulator from its own tracking video, (ii) transfer the generated distribution as a prior to seed a sibling same-species simulator that would otherwise stall, and (iii) thereby populate behaviourally-grounded digital organisms without electrophysiology – the whole-organism analogue of the whole-cell genotype-to-phenotype model (31), of biomechanical movement-simulation platforms (11), and of community whole-animal simulation efforts (55). The F1 NULL ($p=0.734$, §3.13) underwrites the “without electrophysiology” clause: adding internal neural signal to the loss yields no detectable gain. The boundary on this programme is not the optimiser but the simulator’s mechanistic content, which the Jacobian localises (§3.15). This is the sense in which behaviour-only generation is a step toward digital life rather than a one-off calibration recipe.

Read at the level of the field rather than of any one model, the same recipe is a general way past the invasive-measurement bottleneck. Because the formulation is simulator-agnostic and engine-agnostic, and consumes only the externally observable behaviour that ordinary tracking already produces, it could in principle be pointed at any new mechanistic simulator – and, as community models proliferate beyond worm and fly, generating their parameters at scale from behavioural video would no longer require an accompanying physiology study for each one. The reach we find most consequential is to the cases the bottleneck excludes outright. Wherever invasive recording is hard or impossible – species with no historical electrophysiology, endangered animals that cannot be sacrificed for patch-clamp, and preparations that cannot be opened or imaged – behaviour can still, in most cases, be filmed, and a behaviour-only route to calibrated mechanistic models could open those organisms to in-silico study for the first time. We present the worm and fly results as the demonstration that the route works; extending it to such data-poor and hard-to-access species is an outlook this method invites rather than a claim it settles here, but it is, in our view, where the larger payoff lies.

4.2 Behavioural parity and the value-to-default relationship

On the 14-metric behavioural evaluator the generated parameters satisfy the $D_1 < D_2 \ll D_3$ aggregate-distance hierarchy (generated-to-sim-target < generated-to-real $\ll$ random-to-real) across all four simulators; the cross-simulator headline is aggregate behavioural-feature distance to real biology (§3.5). Where a per-metric residual remains, the Jacobian (§3.15) localises it to a specific parameter axis, identifying a simulator mechanism to extend rather than an optimiser failure.

The amount of behavioural ground that generation can close tracks how far each simulator’s deposited default sits from the behaviour the substrate can reach, not the physical richness of the substrate. The statistically-powered PGOB-vs-baseline sweep (§3.5) makes this concrete: where the baseline is an Initialized (randomised) walker (FlyGym v2, 8/10 observables, +32.4%, Holm-significant) there is large room to generate, while where the default is already a good walker (flybody, 0/9 significant) the correct outcome is the measured null – generation rescues Initialized starts and leaves good defaults intact. Crucially, where a genuine real reference exists the generated parameters reach a distance-to-real *comparable* to the simulator’s physiologically-calibrated default (recovered-to-real $\approx$ default-to-real on BAAIWorm; 3.29 vs. 2.69 on modWorm), so the deployment claim is behaviour-only *parity* with physiological calibration – obtained without performing any physiology – not superiority over it.

4.3 Limitations

We state the operating envelope of PGOB precisely, in one place. The binding constraint throughout is the simulator’s mechanistic content, not the optimiser or the loss, and the boundaries below

sharpen that scope rather than qualify the results. *(i) Substrate ceilings are a property of the simulator, not the method.* The modWorm 30-d Cook ODE cannot express the high-order eigenworm modes that distinguish real OWMD worms, so even its expert arm cannot reach those channels and PGOB sits at this ceiling (§3.7); this is a limit of the deposited substrate, and PGOB’s Jacobian localises exactly which mechanism would have to be added. *(ii) Same-species prior transfer uses a biophysically-informed dimension map.* The same-species soft prior maps parameters by biophysical class (gap→gap-junction, synaptic→synaptic, membrane→passive, ion→ion, muscle→motor-output), an informed rather than exact correspondence, and the transfer demonstrated here is between two simulators of the same species (§3.14); whether a comparable prior helps across more distantly related body plans is left open. *(iii) The flybody walk-imitation benchmark is a single MoCap-locked task whose trajectory observables carry no extractable parameter signal,* so actuator perturbations do not propagate to the kinematics (§3.7); this is a property of that benchmark, and the flybody init-mode signal accordingly lives in loss space, where the same-species prior gives a $9.3\times$ gain (§3.8). *(iv) Everything needed to re-derive these boundaries is public:* all code, fixed seeds, and reference data are released for independent reproduction (§Data and code availability).

4.4 Take-home

The contribution is a behaviour-only generation framework: a behaviour-only inverse-problem formulation with a cross-simulator evaluator and a residual-to-mechanism attribution map, into which black-box optimisers and amortised neural posteriors plug interchangeably. PGOB produces whole-organism simulator parameters from a single laboratory’s tracking video under one formulation, evaluator and engine stack across four published simulators spanning two species (only the initialisation is dimension-adaptive; Table 1), and it stands on a chain of held-out tests rather than on a single fitted number: a prospective 43-strain mutant test (leave-one-strain-out $\bar{\rho}=0.684$ vs. strain-shuffle 0.527), a cross-laboratory generalisation test (Schafer↔Brown, left–right phase coupling lab-invariant at $< 3\times$ cross/within W_1), and a same-species prior-transfer test that indicates the generated parameters carry transferable species-specific structure (whether they lie on the biological parameter manifold is left to future work). The residual that generation leaves behind is, in every simulator we examined, a property of the deposited model rather than of the optimiser or the loss: closing it requires extending the simulator itself – a head circuit, the 96-muscle map, or a missing sensory feedback loop – and these are simulator-development moves, not calibration moves. **The binding constraint on whole-organism in-silico neuroscience is the simulator’s mechanistic content, not the optimiser or the loss,** and PGOB’s Jacobian identifies, axis by axis, which mechanism to extend next.

4.5 Future work

(a) Loss-rescuable observables. modWorm undulation phase std is the single loss-rescuable observable in the 4×6 diagnostic (SI §S4.7); a targeted distributional loss on that axis is an immediate seed. **(b) Ephys-augmented PGOB.** flybody walk-imitation default is locked by reference MoCap; augmenting the loss with DANDI leg motor-neuron recordings (when paired walking + per-frame firing become available at $n\geq 10$) would let PGOB move axes the MoCap-locked policy currently fixes. **(c) Bootstrapped digital life and data-poor species.** Composing generation with same-species prior transfer is the natural route to populating behaviourally-grounded digital organisms at scale from tracking video alone, without electrophysiology. The same composition is what could carry the method to organisms that current calibration cannot reach – species with no historical physiology, endangered animals, and preparations that cannot be patched or imaged – for which behaviour

is recordable even when intracellular access is not; closing that gap is, to us, the most valuable direction this framework opens.

Acknowledgements

We thank the BAAIWorm team (72) for releasing the connectome-coupled biomechanical worm; the modWorm authors (50) for the Julia muscle-skeletal model and Cook 2019 connectome assets (8); the Janelia flybody team (62) for the 102-DoF MuJoCo model and walking-imitation deposit (DOI 10.25378/janelia.25309105); the NeuroMechFly/FlyGym team (64); and the OpenWorm Foundation and OpenWorm Movement Database maintainers for the N2 wild-type recordings. Computational resources were provided by cloud GPU compute.

Author contributions

C.H. conceived the framework, implemented the universal evaluator and the cross-simulator drivers, ran all experiments, and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Ethics statement

No ethics approval was required: all analyses use publicly available behavioural datasets without new animal experimentation.

Data and code availability

Code is MIT-licensed at <https://github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior> (tag v1.0-PG0B). All third-party data are public under their original licences (OWMD Zenodo 10.5281/zenodo.5839010; Brown-Tierpsy Zenodo 10.5281/zenodo.3837679; Yemini-2013 Zenodo 10.5281/zenodo.4633435; Janelia walking-imitation Figshare 10.25378/janelia.25309105). The main paper, supplementary information, reproduction code, and figures are archived on Zenodo (DOI 10.5281/zenodo.20691877). Full SHA-256 hashes, sizes, and per-figure reproduction commands are in SI §S6; a consolidated Reproducibility statement is in SI §S6.5.

References

- [1] Diego Aldarondo, Josh Merel, Jesse D. Marshall, Leonard Hasenclever, Ugne Klibaite, Amanda Gellis, Yuval Tassa, Greg Wayne, Matthew Botvinick, and Bence P. Ölveczky. A virtual rodent predicts the structure of neural activity across behaviours. *Nature*, 632(8025):594–602, 2024.
- [2] Anthony Azevedo, Ellen Lesser, Jasper S. Phelps, Brandon Mark, Leila Elabbady, Sumiya Kuroda, et al. Connectomic reconstruction of a female *Drosophila* ventral nerve cord. *Nature*, 631(8020):360–368, 2024.

- [3] Cornelia I. Bargmann and Eve Marder. From the connectome to brain function. *Nature Methods*, 10(6):483–490, 2013.
- [4] Kristin Branson, Alice A. Robie, John Bender, Pietro Perona, and Michael H. Dickinson. High-throughput ethomics in large groups of *Drosophila*. *Nature Methods*, 6(6):451–457, 2009.
- [5] André E. X. Brown, Eviatar I. Yemini, Laura J. Grundy, Tadas Jucikas, and William R. Schafer. A dictionary of behavioral motifs reveals clusters of genes affecting *Caenorhabditis elegans* locomotion. *Proceedings of the National Academy of Sciences*, 110(2):791–796, 2013.
- [6] Vittorio Caggiano, Huawei Wang, Guillaume Durandau, Massimo Sartori, and Vikash Kumar. MyoSuite: a contact-rich simulation suite for musculoskeletal motor control. *Proceedings of Machine Learning Research*, 168:492–507, 2022.
- [7] Mohan Chen, Dazheng Feng, Hongtao Su, Tingwei Su, and Meng Wang. Neural model generating klinotaxis behavior accompanied by a random walk based on *C. elegans* connectome. *Scientific Reports*, 12(1):3043, 2022.
- [8] Steven J Cook, Travis A Jarrell, Christopher A Brittin, et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature*, 571(7763):63–71, 2019.
- [9] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.
- [10] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. *Advances in Neural Information Processing Systems (NeurIPS)*, 26:2292–2300, 2013.
- [11] Scott L. Delp, Frank C. Anderson, Allison S. Arnold, Peter Loan, Ayman Habib, Chand T. John, Eran Guendelman, and Darryl G. Thelen. OpenSim: open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, 54(11):1940–1950, 2007.
- [12] Michael H. Dickinson, Claire T. Farley, Robert J. Full, M. A. R. Koehl, Rodger Kram, and Steven Lehman. How animals move: an integrative view. *Science*, 288(5463):100–106, 2000.
- [13] Sven Dorkenwald, Arie Matsliah, Amy R. Sterling, et al. Neuronal wiring diagram of an adult brain. *Nature*, 634:124–138, 2024.
- [14] Conor Durkan, Artur Bekasov, Iain Murray, and George Papamakarios. Neural spline flows. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [15] Anthony D. Fouad, Shelly Teng, Julian R. Mark, Alice Liu, Pilar Alvarez-Illera, Hongfei Ji, et al. Distributed rhythm generators underlie *Caenorhabditis elegans* forward locomotion. *eLife*, 7:e29913, 2018.
- [16] Shangbang Gao, Sihui Asuka Guan, Anthony D. Fouad, Jun Meng, Taizo Kawano, Yung-Chi Huang, et al. Excitatory motor neurons are local oscillators for backward locomotion. *eLife*, 7:e29915, 2018.
- [17] Pdraig Gleeson, David Lung, Radu Grober, et al. c302: a multiscale framework for modelling the nervous system of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170379, 2018.

- [18] Pedro J Gonçalves, Jan-Matthis Lueckmann, Michael Deistler, et al. Training deep neural density estimators to identify mechanistic models of neural dynamics. *eLife*, 9:e56261, 2020.
- [19] Jesse M. Gray, Joseph J. Hill, and Cornelia I. Bargmann. A circuit for navigation in *Caenorhabditis elegans*. *PNAS*, 102:3184–3191, 2005.
- [20] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1861–1870, 2018.
- [21] Nikolaus Hansen. The CMA evolution strategy: A tutorial. *arXiv preprint arXiv:1604.00772*, 2016.
- [22] Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9(2):159–195, 2001.
- [23] Michael L. Hines and Nicholas T. Carnevale. The NEURON simulation environment. *Neural Computation*, 9(6):1179–1209, 1997.
- [24] Alan L. Hodgkin and Andrew F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4):500–544, 1952.
- [25] Auke Jan Ijspeert, Alessandro Crespi, Dimitri Ryczko, and Jean-Marie Cabelguen. From swimming to walking with a salamander robot driven by a spinal cord model. *Science*, 315(5817):1416–1420, 2007.
- [26] Eduardo J. Izquierdo and Randall D. Beer. Connecting a connectome to behavior: an ensemble of neuroanatomical models of *C. elegans* klinotaxis. *PLoS Computational Biology*, 9(2):e1002890, 2013.
- [27] Eduardo J Izquierdo and Randall D Beer. Connecting a connectome to behavior: an ensemble of neuroanatomical models of *C. elegans* klinotaxis. *PLoS Computational Biology*, 14(2):e1005968, 2018.
- [28] Avelino Javier, Michael Currie, Chee Wai Lee, Jim Hokanson, Kezhi Li, Céline N. Martineau, Eviatar Yemini, Laura J. Grundy, Chris Li, QueeLim Ch’ng, William R. Schafer, Ellen A. A. Nollen, Rex Kerr, and André E. X. Brown. An open-source platform for analyzing and sharing worm-behavior data. *Nature Methods*, 15(9):645–646, 2018.
- [29] Pierre Karashchuk, Katie I. Rupp, Evyn S. Dickinson, Sarah Walling-Bell, Elischa Sanders, Eiman Azim, Bingni W. Brunton, and John C. Tuthill. Anipose: a toolkit for robust markerless 3D pose estimation. *Cell Reports*, 36(13):109730, 2021.
- [30] Jan Karbowski. Deciphering neural circuits for *Caenorhabditis elegans* behavior by computations and perturbations to genome and connectome. *Current Opinion in Systems Biology*, 13:44–51, 2019.
- [31] Jonathan R. Karr, Jayodita C. Sanghvi, Derek N. Macklin, Miriam V. Gutschow, Jared M. Jacobs, Benjamin Bolival Jr, Nacyra Assad-Garcia, John I. Glass, and Markus W. Covert. A whole-cell computational model predicts phenotype from genotype. *Cell*, 150(2):389–401, 2012.

- [32] Saul Kato, Harris S Kaplan, Tina Schrödel, et al. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell*, 163(3):656–669, 2015.
- [33] Janne K. Lappalainen, Fabian D. Tschopp, Sridhama Prakhya, Mason McGill, Aljoscha Nern, Kazunori Shinomiya, Shin-ya Takemura, Eyal Gruntman, Jakob H. Macke, and Srinivas C. Turaga. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634:1132–1140, 2024.
- [34] Victor Lobato-Rios, Shravan Tata Ramalingasetty, Pembe Gizem Özdil, Jonathan Arreguit, Auke Jan Ijspeert, and Pavan Ramdya. NeuroMechFly, a neuromechanical model of adult *Drosophila melanogaster*. *Nature Methods*, 19(5):620–627, 2022.
- [35] Johan M. Melis, Igor Siwanowicz, and Michael H. Dickinson. Machine learning reveals the control mechanics of an insect wing hinge. *Nature*, 628(8009):795–803, 2024.
- [36] Josh Merel, Diego Aldarondo, Jesse Marshall, Yuval Tassa, Greg Wayne, and Bence Ölveczky. Deep neuroethology of a virtual rodent. *arXiv preprint arXiv:1911.09451*, 2019. International Conference on Learning Representations (ICLR) 2020.
- [37] Florian T. Muijres, Michael J. Elzinga, Johan M. Melis, and Michael H. Dickinson. Flies evade looming targets by executing rapid visually directed banked turns. *Science*, 344(6180):172–177, 2014.
- [38] Jeffrey P. Nguyen, Frederick B. Shipley, Ashley N. Linder, George S. Plummer, Mochi Liu, Sagar U. Setru, Joshua W. Shaevitz, and Andrew M. Leifer. Whole-brain calcium imaging with cellular resolution in freely behaving *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences*, 113(8):E1074–E1081, 2016.
- [39] George Papamakarios and Iain Murray. Fast ε -free inference of simulation models with Bayesian conditional density estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [40] George Papamakarios, Theo Pavlakou, and Iain Murray. Masked autoregressive flow for density estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [41] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. DeepMimic: example-guided deep reinforcement learning of physics-based character skills. *ACM Transactions on Graphics*, 37(4):143, 2018.
- [42] Talmo D. Pereira, Joshua W. Shaevitz, and Mala Murthy. Quantifying behavior to understand the brain. *Nature Neuroscience*, 23(12):1537–1549, 2020.
- [43] Talmo D. Pereira, Nathaniel Tabris, Arie Matsliah, David M. Turner, Junyu Li, Shruthi Ravindranath, et al. SLEAP: a deep learning system for multi-animal pose tracking. *Nature Methods*, 19(4):486–495, 2022.
- [44] Gabriel Peyré and Marco Cuturi. Computational optimal transport: With applications to data science. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- [45] Francesco Randi and Andrew M Leifer. Multi-modal whole-organism neuroscience: Integrating behavior, neural dynamics, and physiology in *C. elegans*. *Current Opinion in Neurobiology*, 79: 102684, 2023.

- [46] Andrea Saltelli, Marco Ratto, Terry Andres, Francesca Campolongo, Jessica Cariboni, Debora Gatelli, Michaela Saisana, and Stefano Tarantola. *Global Sensitivity Analysis: The Primer*. John Wiley & Sons, 2008.
- [47] Sanjay P. Sane. The aerodynamics of insect flight. *Journal of Experimental Biology*, 206(23): 4191–4208, 2003.
- [48] Gopal P. Sarma, Chee Wai Lee, Tom Portegys, Vahid Ghayoomie, Travis Jacobs, Bradly Alicea, et al. OpenWorm: overview and recent advances in integrative biological simulation of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170382, 2018.
- [49] Louis K. Scheffer, C. Shan Xu, Michal Januszewski, et al. A connectome and analysis of the adult *Drosophila* central brain. *eLife*, 9:e57443, 2020.
- [50] Eli Shlizerman et al. modworm: a modular muscular-skeletal c. elegans simulator. *Preprint*, 2025.
- [51] James C Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992.
- [52] James C. Spall. *Introduction to Stochastic Search and Optimization: Estimation, Simulation, and Control*. John Wiley & Sons, Hoboken, NJ, 2003.
- [53] Greg J Stephens, Bethany Johnson-Kerner, William Bialek, and William S Ryu. Dimensionality and dynamics in the behavior of *C. elegans*. *PLoS Computational Biology*, 4(4):e1000028, 2008.
- [54] Greg J. Stephens, Matthew Bueno de Mesquita, William S. Ryu, and William Bialek. Emergence of long timescales and stereotyped behaviors in *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences*, 108(18):7286–7289, 2011.
- [55] Balázs Szigeti, Pádraig Gleeson, Michael Vella, et al. OpenWorm: An open-science approach to modeling *Caenorhabditis elegans*. *Frontiers in Computational Neuroscience*, 8:137, 2014.
- [56] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind control suite. *arXiv preprint arXiv:1801.00690*, 2018.
- [57] Alvaro Tejero-Cantero, Jan Boelts, Michael Deistler, Jan-Matthis Lueckmann, Conor Durkan, Pedro J Gonçalves, David S Greenberg, and Jakob H Macke. sbi: A toolkit for simulation-based inference. *Journal of Open Source Software*, 5(52):2505, 2020.
- [58] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.
- [59] Saran Tunyasuvunakool, Alistair Muldal, Yotam Doron, Siqi Liu, Steven Bohez, Josh Merel, Tom Erez, Timothy Lillicrap, Nicolas Heess, and Yuval Tassa. dm_control: software and tasks for continuous control. *Software Impacts*, 6:100022, 2020.
- [60] Anne E. Urai, Brent Doiron, Andrew M. Leifer, and Anne K. Churchland. Large-scale neural recordings call for new insights to link brain and behavior. *Nature Neuroscience*, 25(1):11–19, 2022.

- [61] Lav R. Varshney, Beth L. Chen, Eric Paniagua, David H. Hall, and Dmitri B. Chklovskii. Structural properties of the *Caenorhabditis elegans* neuronal network. *PLoS Computational Biology*, 7(2):e1001066, 2011.
- [62] Roman Vaxenburg, Igor Siwanowicz, Josh Merel, Alice A Robie, Carmen Morrow, Guido Novati, Sam Stupski, Gwyneth M Card, and Srinivas C Turaga. Whole-body simulation of *drosophila* reveals motor control of behavior. *Nature*, 2025. flybody release, Figshare DOI 10.25378/janelia.25309105.
- [63] Alejandro F. Villaverde and Julio R. Banga. Reverse engineering and identification in systems biology: strategies, perspectives and challenges. *Journal of the Royal Society Interface*, 11(91):20130505, 2014.
- [64] Sibowang-Chen et al. Neuromechfly v2: simulating embodied sensorimotor control in adult *drosophila*. *Nature Methods*, 2024.
- [65] Quan Wen, Michelle D. Po, Elizabeth Hulme, Sway Chen, Xinyu Liu, Sze Wai Kwok, et al. Proprioceptive coupling within motor neurons drives *C. elegans* forward locomotion. *Neuron*, 76(4):750–761, 2012.
- [66] Quan Wen, Shangbang Gao, and Mei Zhen. *Caenorhabditis elegans* excitatory ventral cord motor neurons derive rhythm for body undulation. *Philosophical Transactions of the Royal Society B*, 373(1758):20170370, 2018.
- [67] John G White, Eileen Southgate, J Nichol Thomson, and Sydney Brenner. The structure of the nervous system of the nematode *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society of London B*, 314(1165):1–340, 1986.
- [68] Daniel Witvliet, Ben Mulcahy, James K. Mitchell, Yaron Meirovitch, Daniel R. Berger, Yuelong Wu, et al. Connectomes across development reveal principles of brain maturation. *Nature*, 596(7871):257–261, 2021.
- [69] Gang Yan, Petra E. Vértés, Emma K. Towlson, Yee Lian Chew, Denise S. Walker, William R. Schafer, and Albert-László Barabási. Network control principles predict neuron function in the *Caenorhabditis elegans* connectome. *Nature*, 550(7677):519–523, 2017.
- [70] Eviatar Yemini, Tadas Jucikas, Laura J. Grundy, André E. X. Brown, and William R. Schafer. A database of *Caenorhabditis elegans* behavioral phenotypes. *Nature Methods*, 10:877–879, 2013.
- [71] Anthony Zador, Sean Escola, Blake Richards, Bence Ölveczky, Yoshua Bengio, Kwabena Boahen, et al. Catalyzing next-generation artificial intelligence through NeuroAI. *Nature Communications*, 14:1597, 2023.
- [72] Mengdi Zhao, Ning Wang, Xinyu Jiang, Xiaohan Ma, Haoran Ma, Gan He, Kai Du, Lei Ma, and Tiejun Huang. An integrative data-driven model simulating *C. elegans* brain, body and environment interactions. *Nature Computational Science*, 4:106–120, 2024.